

Changes in Travel Behaviour to Education and Work in New Zealand: A Comparative Analysis of Census 2018 and 2023 Data

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Abstract

This study examines how travel behaviour to education and work changed in New Zealand between the 2018 and 2023 population censuses, with particular attention to whether post-COVID shifts differed systematically between journey purposes. Using nationally aggregated census data from Statistics New Zealand, Journey to Education and Journey to Work datasets were harmonised under consistent travel-mode classifications (At Home, Private Transport, Public Transport, Active Transport, and Other Transport) to enable direct comparison across time, geography, and demographic subgroups.

The analysis combines descriptive mode-share comparisons, percentage-point change estimates, formal difference-in-differences calculations, Pearson chi-square tests, and a multinomial logistic regression model incorporating a $Year \times JourneyType$ interaction. Results reveal statistically significant redistribution in travel-mode shares between 2018 and 2023 for both journey purposes. The most pronounced structural shift is the substantial increase in At Home participation, alongside persistent declines in public transport use. Interaction testing confirms that the magnitude and direction of change differ significantly between education and work travel, indicating heterogeneous post-pandemic adjustment patterns.

Regional and demographic extensions demonstrate marked spatial and age-based variation. Urban regions exhibit stronger modal restructuring than non-urban areas, and divergence between education and work becomes more pronounced among adult and older cohorts. Overall, the findings provide robust national-level evidence of uneven post-pandemic mobility restructuring in New Zealand, with implications for transport planning, public transport recovery strategies, and long-term infrastructure investment under evolving work and study practices.

1 Introduction & Motivation

Understanding how people travel to education and work is central to transport planning and public policy, as these journeys account for a substantial share of daily travel demand and strongly influence peak-period congestion, infrastructure investment, environmental outcomes, and social equity. In New Zealand, population-level travel behaviour is most comprehensively captured through the national census, which provides consistent, spatially detailed information on individuals’ main means of travel across regions, districts, and neighbourhood-scale statistical areas. Census data therefore play a critical role in informing long-term transport strategies and monitoring structural changes in travel behaviour over time (Statistics New Zealand, 2024a). The 2018 and 2023 census waves are of particular analytical significance because they bracket the COVID-19 pandemic, a period that fundamentally disrupted everyday mobility patterns. Lockdowns, school closures, remote learning, and the rapid adoption of working and studying from home altered not only the volume of travel but also the relative attractiveness and feasibility of different travel modes (G. Currie *et al.*, 2021; K. Pawluk De-Toledo *et al.*, 2024). While early pandemic impacts were widely documented, there remains considerable uncertainty regarding the extent to which these changes represent temporary shocks versus persistent structural shifts in travel behaviour, especially in a post-pandemic context where hybrid arrangements have become increasingly normalised (D. A. Hensher *et al.*, 2022; Z. Huang *et al.*, 2023).

Most empirical research on post-COVID travel behaviour has focused on journeys to work, reflecting their historical dominance in transport modelling and policy evaluation (E. Heinen *et al.*, 2015). However, travel to education represents a distinct and substantial component of daily travel demand, involving different age groups, household decision-making processes, and institutional constraints (S. Mandic *et al.*, 2023). In New Zealand, Journey to Education data are only available for the 2018 and 2023 censuses, making this the first opportunity to conduct a national pre- and post-COVID comparison of education-related travel behaviour using consistent population-scale data (Statistics New Zealand, 2024b). Despite growing international evidence on pandemic-induced mobility change, no national-scale study has directly compared post-COVID changes in Journey to Education and Journey to Work travel-mode distributions in New Zealand using harmonised census data within a unified statistical framework. Existing analyses tend either to focus on commuting alone or to rely on survey-based evidence from specific cities (R. Buehler *et al.*, 2025). This leaves an important gap in understanding whether education and work travel have followed similar or divergent recovery trajectories across regions and settlement types.

Preliminary descriptive evidence suggests that travel behaviour changes since 2018 have not been uniform across journey purposes or regions. Study from home has increased markedly, particularly for education, while public transport has experienced sustained declines that vary by urbanisation level and geographic context (D. A. Hensher *et al.*, 2022; R. Buehler *et al.*, 2025). At the same time, private transport continues to dominate in-person travel, and changes in active transport appear uneven across cities and regions (K. Pawluk De-Toledo *et al.*, 2024). These patterns raise important questions for transport policy, including whether post-COVID travel behaviour is converging toward more sustainable outcomes or reinforcing existing car dependence, and whether education and work travel are evolving along similar or divergent trajectories.

This research builds directly on prior internship work undertaken with the New Zealand Transport Agency (NZTA), where Census 2018 and 2023 Journey to Education data were analysed and presented through an interactive Power BI dashboard. That work highlighted clear national and regional shifts in travel-mode shares but was primarily descriptive. The present study extends this

foundation by integrating Journey to Education and Journey to Work data within a unified analytical framework and applying formal statistical methods to assess whether observed differences over time and between journey purposes are statistically meaningful rather than visually apparent alone.

Accordingly, this study examines (i) whether travel-mode distributions changed between 2018 and 2023 within Journey to Education and Journey to Work; (ii) whether travel-mode distributions differ between Education and Work journeys within each census year; and (iii) whether the magnitude and direction of time-based changes differ systematically between journey purposes. It further evaluates whether these differences exhibit spatial heterogeneity across urban and non-urban contexts and whether differential change over time can be formally detected when census year and journey purpose are modelled jointly within a multinomial framework. These questions are addressed using harmonised mode categories, percentage-point and difference-in-differences comparisons, chi-square tests of homogeneity and independence, and a multinomial specification incorporating a $Year \times Journey$ Type interaction term.

To address these questions, the study adopts a comparative distributional framework suitable for aggregated census data. The analysis focuses on system-level redistribution of travel modes rather than individual-level behavioural causation, making distributional comparisons and multinomial modelling appropriate for detecting structural shifts across time and journey purposes.

By combining population-scale census data with formal statistical inference and complementary visual analytics, this study provides robust national-level evidence on post-COVID travel behaviour in New Zealand. In doing so, it addresses a clear empirical gap regarding education travel, offers the first harmonised national comparison of education and work mode-share change, and contributes policy-relevant insight into how different segments of the transport system have adapted in the post-pandemic environment.

2 Background and Literature Review

This section situates the study within the broader literature on travel behaviour change, with particular emphasis on journey purpose, spatial heterogeneity, and pandemic-induced disruption. It first reviews established research on mode choice and routine mobility patterns, before examining COVID-19 as a natural experiment in travel behaviour change. The review then considers evidence specific to education travel and regional variation, highlighting limitations in existing work and motivating the comparative, population-scale framework adopted in this study.

2.1 Travel behaviour change as a long-standing research domain

Research on travel behaviour has historically concentrated on explaining mode choice for routine journeys, particularly journeys to work, reflecting their importance for peak-period congestion, infrastructure provision, and labour market accessibility. This body of literature has employed household travel surveys, census data, and discrete choice models to examine how socio-demographic characteristics, land use patterns, and transport infrastructure shape travel decisions (E. Heinen *et al.*, 2015). Across this work, travel behaviour is consistently shown to be structured by spatial context, daily activity constraints, and the availability of viable transport alternatives.

Within this literature, journeys to work have received disproportionate attention relative to journeys to education. The dominance of commuting-based modelling has implicitly positioned work

travel as the default representation of daily mobility, potentially obscuring important differences in education-related travel demand. Studies focusing on school and education travel demonstrate that distance, settlement type, parental perceptions, and infrastructure availability strongly influence mode choice, often producing different patterns from adult commuting (S. Mandic *et al.*, 2023; Y. Yan *et al.*, 2019). These structural differences suggest that findings derived from Journey to Work analyses cannot be assumed to generalise automatically to Journey to Education contexts.

2.2 COVID-19 as a natural experiment in travel behaviour change

The COVID-19 pandemic represents an unprecedented disruption to daily mobility, widely conceptualised as a global natural experiment in travel behaviour change. A rapidly expanding literature documents dramatic declines in overall travel demand during lockdown periods, alongside sharp shifts in mode shares away from public transport and towards private vehicles, active modes, and remote participation (G. Currie *et al.*, 2021; K. Pawluk De-Toledo *et al.*, 2024). Unlike incremental policy or infrastructure interventions, the pandemic simultaneously altered perceived health risk, institutional arrangements, and technological feasibility, particularly through the widespread adoption of working and studying from home.

Evidence consistently identifies working from home (WFH) as a central driver of reduced commuting demand during the pandemic, with hybrid work arrangements persisting beyond the acute phase (D. A. Hensher *et al.*, 2022; Z. Huang *et al.*, 2023). These findings suggest that COVID-induced behavioural changes may represent structural adjustments rather than temporary deviations, with emerging longitudinal evidence documenting persistent post-pandemic travel shifts (K. Hamad *et al.*, 2024). Public transport has experienced the most severe and prolonged decline, influenced by infection risk perceptions and service disruptions, while private vehicle use rebounded more rapidly (R. Buehler *et al.*, 2025; M. S. Hossain *et al.*, 2024). Active transport exhibited heterogeneous responses across cities and contexts (K. Pawluk De-Toledo *et al.*, 2024).

This asymmetric recovery raises the possibility that different journey purposes may exhibit distinct post-pandemic trajectories, particularly where flexibility, institutional constraints, and household decision-making processes differ.

2.3 Education travel and differential impacts of the pandemic

Although much COVID-related travel research focuses on adult commuting, a smaller but growing literature examines travel to school and education, revealing both parallels and divergences. New Zealand evidence shows that school travel mode choice varies strongly by settlement type and home-to-school distance, with active travel more prevalent in compact urban settings (S. Mandic *et al.*, 2023). Pandemic-related disruptions may therefore manifest differently across spatial contexts.

Moreover, education travel decisions are frequently mediated through household coordination and institutional rules rather than individual choice alone. Prior research has documented substantial differences in school travel mode choice between children and youth, highlighting age-specific behavioural dynamics within education-related mobility (R. Mitra *et al.*, 2010). Research indicates that COVID-19 altered adolescents’ attitudes and parental perceptions regarding active travel (S. Mandic *et al.*, 2024), suggesting that education travel responses may be shaped by factors distinct from those influencing adult commuting. International evidence further demonstrates that institutional context, including school type and governance, can produce systematic differences in travel behaviour even within the same metropolitan region (Y. Yan *et al.*, 2019).

Taken together, this literature underscores the importance of treating education travel as analytically distinct rather than extrapolating from work-focused studies.

2.4 Persistence, recovery, and regional heterogeneity

A key unresolved question concerns whether pandemic-induced travel behaviour changes are temporary or persistent. More recent studies incorporating extended post-pandemic observation windows show uneven recovery patterns, particularly for public transport, with substantial variation across regions and modes (R. Buehler *et al.*, 2025). Qualitative research suggests that behavioural adjustments are embedded within broader lifestyle changes, including altered expectations of flexibility and decentralised activity patterns (K. Pawluk De-Toledo *et al.*, 2024).

Despite these advances, much of the existing evidence remains survey-based or city-specific. There is comparatively limited use of population-scale administrative data to compare changes consistently across journey purposes and spatial contexts. In New Zealand, census data provide rare national coverage for both Journey to Work and Journey to Education (Statistics New Zealand, 2024a), yet systematic comparative analysis remains limited.

2.5 Methodological approaches and relevance to the present study

Methodologically, prior research employs multinomial logit models, quasi-experimental designs, and before-after comparisons to analyse travel behaviour change (E. Heinen *et al.*, 2015; D. A. Hensher *et al.*, 2022; Y. Deneke *et al.*, 2024). Where interest centres on redistribution across categorical outcomes, distributional approaches are particularly appropriate. Chi-square tests of homogeneity and independence provide a natural framework for assessing whether multinomial travel-mode distributions differ across time periods or journey types in large contingency tables (L. A. Goodman, 1982).

Given the aggregated structure of census data and the study’s focus on system-level redistribution rather than individual-level causal inference, a distributional framework is both statistically coherent and substantively meaningful. Percentage-point and relative changes complement formal significance testing, helping to distinguish statistically detectable differences from substantively important shifts in large samples. The inclusion of a multinomial modelling framework with a $Year \times JourneyType$ interaction allows direct testing of differential change within a unified model structure.

By integrating national census data comprising over one million aggregated records across Journey to Education and Journey to Work for the 2018 and 2023 census waves, spanning all regions and statistical areas in New Zealand, this study responds to calls for population-scale evidence and greater attention to spatial heterogeneity (S. Mandic *et al.*, 2023; R. Buehler *et al.*, 2025). The combination of inferential statistics and visual analytic techniques further enhances interpretability for policy audiences (K. Pawluk De-Toledo *et al.*, 2024).

2.6 Contribution of the current research

This study makes four principal contributions.

First, it provides the first harmonised national comparison of Journey to Education and Journey to Work travel-mode distributions before and after the COVID-19 pandemic in New Zealand.

Second, it moves beyond descriptive analysis by formally testing differential change over time using distributional methods and multinomial modelling.

Third, it explicitly examines spatial heterogeneity across urban and non-urban contexts, addressing an underexplored dimension of post-pandemic travel recovery.

Fourth, it grounds post-COVID mobility analysis in comprehensive population-scale census data rather than survey-based stated preferences alone.

Together, these contributions advance understanding of how different segments of the transport system have adapted in the post-pandemic period and provide robust empirical evidence to inform long-term transport planning and investment decisions.

3 Methodology

This section outlines the data and analytical framework used to examine changes in travel-mode distributions between the 2018 and 2023 Censuses. Section 3.1 describes the data sources, download procedures, and preparation steps, including harmonisation and confidentiality-aware processing. Section 3.2 defines the descriptive measures of mode composition and temporal change, while Section 3.3 introduces the Chi-square framework for testing distributional homogeneity and effect size assessment. Section 3.4 presents the multinomial interaction model used to evaluate differential change between journey types. Section 3.5 extends the analysis to regional, urban-non-urban, city-level, and age-stratified comparisons, followed by documentation of software and reproducibility.

3.1 Data collection, preparation, and governance

This study employs a structured, reproducible data pipeline to examine changes in travel behaviour to education and work across New Zealand using national Census data (Statistics New Zealand, 2024a). The methodology was designed to ensure aggregation safety, cross-dataset comparability, and compliance with confidentiality, privacy, and Māori data governance requirements. All data processing steps were applied exclusively to published, aggregated Census counts and are fully documented to support transparency and reproducibility.

3.1.1 Data sources and datasets

The analysis draws on publicly available, aggregated Census data released by Statistics New Zealand (Stats NZ) through the Census Transport data catalogue. Two datasets were selected to represent the primary journey purposes of interest:

- **Main means of travel to education, age, and gender for the census usually resident population who are studying (2018 and 2023 Censuses)** (Statistics New Zealand, 2024b)
- **Main means of travel to work, total personal income, age, and gender for the employed census usually resident population aged 15 years and over (2018 and 2023 Censuses)** (Statistics New Zealand, 2024c)

The education dataset represents the census usually resident population who reported studying, while the work dataset represents the employed usually resident population aged 15 years and over. Both datasets provide aggregated counts disaggregated by travel mode, age group, gender, Census year, and geographic unit.

Data are available at multiple spatial resolutions. For consistency and aggregation safety, data were downloaded and processed at the **Statistical Area 2 (SA2)** level, which provides stable sub-regional units of approximately 2,000-4,000 residents and ensures comparability across census waves. SA2-level data were subsequently aggregated to regional, territorial authority (city/district), and grouped urban/non-urban classifications for analysis and reporting. Regional council and territorial authority identifiers were retained to support transparent spatial aggregation and comparative analysis.

All datasets consist exclusively of aggregated population counts. No unit-record or individual-level data were accessed.

3.1.2 Data access and download procedure

Data were accessed via the Stats NZ online Aotearoa Data Explorer under the Census *Transport* category. For each journey type, Journey to Education and Journey to Work, data for the **2018 and 2023 Census waves were downloaded simultaneously** to ensure consistent table structure and metadata alignment across years.

Data were extracted at the SA2 level using the option “*Total - New Zealand by territorial authority and Auckland local board/SA2*”. For each territorial authority or city, all associated SA2 units were selected and downloaded as separate CSV files. This approach preserved the full national spatial coverage while allowing scalable ingestion and processing.

Downloaded files were named using a consistent <Region>_<District> convention. Region and district identifiers were subsequently parsed from file names and added explicitly as variables during ingestion, ensuring traceability between source files and processed records.

The Journey to Education dataset was downloaded in **November 2025**, and the Journey to Work dataset in **January 2026**. These access dates are reported to support reproducibility, acknowledging that classifications and metadata may evolve over time.

3.1.3 Download filters and selection criteria

Careful filter selection was applied during data extraction to ensure that all counts were **mutually exclusive, aggregation-safe, and analytically meaningful**. In particular, total and residual categories that would generate duplicate rows or inflate counts during aggregation were systematically excluded.

3.1.3.1 Journey to Education dataset

For the Journey to Education dataset (Statistics New Zealand, 2024b):

- Under **Age**, the categories “*Total - age*” and “*Median - age*” were excluded. Broad age-group categories relevant to education participation were retained, while fine-grained sub-groups were excluded to avoid unnecessary fragmentation.

- Under **Main means of travel to education**, the categories “*Total - main means of travel to education*”, “*Total stated - main means of travel to education*”, and “*Not elsewhere included*” were excluded.
- Under **Gender**, the category “*Total - gender*” was retained alongside disaggregated gender categories to enable reconstruction of suppressed values during data preparation.

3.1.3.2 Journey to Work dataset

For the Journey to Work dataset (Statistics New Zealand, 2024c):

- Under **Census year**, the 2013 Census was excluded; only the 2018 and 2023 waves were retained.
- Under **Total personal income**, only the aggregate “*Total - total personal income*” category was selected, with all income sub-categories excluded, as income was not used in subsequent analyses.
- Under **Age**, the categories “*Total - age*” and “*Median - age*” were excluded.
- Under **Main means of travel to work**, the categories “*Total - main means of travel*”, “*Total stated - main means of travel*”, and “*Not elsewhere included*” were excluded.
- Under **Gender**, the “*Total - gender*” category was retained to support reconstruction of suppressed gender values.

Across both datasets, total-level filter options were excluded wherever they would introduce duplicated aggregates. This ensured that summing across categories reproduced published totals exactly and avoided artificial inflation of counts.

3.1.4 Data ingestion, preparation, and harmonisation

As the Census transport tables are released exclusively as **aggregated counts** and are subject to confidentiality protections, data preparation focused on ensuring structural comparability, aggregation validity, and consistent categorisation across datasets and Census waves. The preparation process was designed to preserve the integrity of published totals while enabling robust descriptive and inferential analysis at fine geographic resolution.

Across both Journey to Education and Journey to Work datasets, preparation followed a consistent sequence: consolidation of regional extracts, controlled reconstruction of suppressed values, transformation into a harmonised analytical structure, and verification of aggregation safety prior to analysis. All operations were applied deterministically to published aggregate data, and no statistical imputation or model-based estimation was performed.

3.1.4.1 Consolidation of regional Census extracts

Both datasets were provided as multiple Census extracts, each corresponding to a specific Region-District selection. These files were consolidated into single national datasets to support systematic analysis. During consolidation, region and district identifiers were retained explicitly for every observation, ensuring that all records remained traceable to their original geographic context and enabling transparent aggregation at multiple spatial scales.

Non-analytical structural metadata included in the Census dissemination tables were removed to retain only the variables required for analysis, namely Census year, geographic identifiers, travel

mode, age group, gender category, and population count. This ensured a consistent and minimal schema across all records.

Where structurally valid category combinations contained no recorded observations, counts were encoded as zero. This treatment reflects the interpretation that absence of recorded counts within a published category represents zero contribution rather than missing information, and it ensures numerical stability in subsequent aggregation and contingency-table construction.

3.1.4.2 Handling of confidentiality and reconstruction of gender counts

Statistics New Zealand applies confidentiality protection to Census tables through suppression and controlled adjustment, which can result in missing values at detailed disaggregation levels. In both datasets, a published **Total - gender** category is available within each Census grouping. This total was used as a constraint to reconstruct suppressed gender values where necessary, while preserving all published values.

For the Journey to Education dataset, suppressed values in the “Another gender” category were reconstructed using the identity $\text{Another} = \text{Total} - \text{Male} - \text{Female}$, applied within internally consistent groupings defined by Census year, SA2, travel mode, and age group. Reconstructed values were constrained to be non-negative to avoid artefacts arising from rounding or suppression. Following reconstruction, all Total - gender rows were removed to prevent duplication during aggregation.

The Journey to Work dataset required a more selective approach, as confidentiality suppression may apply to specific gender categories while others remain published. Reconstruction was therefore restricted to the suppressed category only, using the published total and the remaining observed gender counts as constraints. Published values were retained unchanged wherever they were available. As with the education data, Total - gender rows were removed once reconstruction was complete.

This targeted strategy ensures internal consistency with published totals while avoiding distortion of valid, released counts. Importantly, reconstruction was purely arithmetic and deterministic, relying solely on information already present in the Census tables.

3.1.4.3 Harmonisation of analytical structure

Following consolidation and confidentiality handling, both datasets were transformed into a common analytical structure to enable direct comparison between education and work journeys.

Variable names and labels were standardised across datasets, and age-group and gender labels were harmonised to ensure consistency. Geographic labels were normalised to resolve encoding inconsistencies that could otherwise generate artificial duplication of regions during aggregation.

Detailed travel-mode categories differ slightly between education and work Census tables. To establish a stable and policy-relevant basis for comparison, all modes were recoded into five harmonised categories used consistently throughout the analysis:

- **Private Transport**
- **Public Transport**
- **Active Transport**
- **At Home**
- **Other Transport**

This categorisation preserves the key behavioural distinctions relevant to transport policy while ensuring that education and work journeys are directly comparable across Census years.

Any remaining structurally valid but unobserved category combinations were encoded as zero counts. This ensures that later aggregation, proportion calculations, and inferential tests operate on complete and coherent contingency structures. This differs from confidentiality suppression, which is flagged and reconstructed using totals.

A journey-type indicator was added to distinguish education from work journeys, enabling pooled analysis and interaction-based comparisons in later stages.

3.1.4.4 Final integration and validation

The harmonised education and work datasets were combined into a single analytical dataset with a shared schema. Prior to analysis, validation checks confirmed consistency of variable structure, complete coverage across journey types and Census years, and absence of missing values in key analytical fields such as travel category and population count.

These checks provide assurance that observed differences in travel behaviour identified in subsequent sections reflect genuine differences in Census-reported distributions rather than artefacts introduced during data preparation.

The Table 1 below verifies that each harmonised travel category is present for both education and work journeys in the 2018 and 2023 Censuses, confirming structural consistency of the merged dataset.

Table 1: Structural validation of harmonised travel categories across journey types and Census years in the merged analytical dataset

Journey Type	Travel Category	Census Years Represented
Education	Active Transport	2018, 2023
Education	At Home	2018, 2023
Education	Other Transport	2018, 2023
Education	Private Transport	2018, 2023
Education	Public Transport	2018, 2023
Work	Active Transport	2018, 2023
Work	At Home	2018, 2023
Work	Other Transport	2018, 2023
Work	Private Transport	2018, 2023
Work	Public Transport	2018, 2023

3.1.5 Ethics, privacy, and Māori data considerations

This study uses secondary data that are publicly released by Stats NZ in aggregated, anonymised form. All data were provided in accordance with the *Data and Statistics Act 2022* and the *Privacy Act 2020*, with confidentiality protection applied by Stats NZ prior to release. No personal, identifiable, or unit-record data were accessed, and re-identification of individuals is not possible.

As the analysis is based solely on aggregated Census counts, formal human ethics approval was not required.

The datasets include population counts for Māori and non-Māori individuals as part of the national Census. In line with Stats NZ guidance and the principles of the *Māori Data Governance Model*,

Māori data are analysed only in their published aggregate form. No inference is made at the level of individuals, whānau, hapū, or iwi, and no Māori-specific classifications are altered or redefined.

Gender categories were originally provided using bilingual English-te reo Māori labels. For analytical clarity and consistency across datasets, these labels were standardised during preparation. This involved label harmonisation only and did not alter underlying published counts.

3.2 Data Analysis - Analytical Methods and Statistical Framework

This section outlines the quantitative framework used to evaluate national-level changes in travel behaviour between the 2018 and 2023 Censuses, and to compare the evolution of Journey to Education and Journey to Work over time. The analytical approach proceeds in three stages:

1. Estimation of national travel mode composition
2. Measurement of percentage-point change between Census waves
3. Difference-in-differences comparison between journey types

All measures are derived from aggregated Census counts and represent population-level quantities.

3.2.1 National-Level Travel Mode Composition

To examine how travel behaviour is distributed across transport categories, national-level mode shares are first calculated separately for education and work journeys. Because the Census provides counts at the SA2 level, national totals are obtained by summing across all SA2 units. These totals are exact population counts and do not involve sampling or estimation.

Let

$$N_{j,y,c}$$

denote the total number of individuals for journey type

$j \in \{\text{Education, Work}\},$

census year

$y \in \{2018, 2023\},$

and travel category

$c \in \{\text{Private Transport, Public Transport, Active Transport, At Home, Other Transport}\}.$

The total number of journeys for journey type j in census year y is defined as:

$$N_{j,y} = \sum_c N_{j,y,c}$$

The national-level **mode share** for category c is then given by:

$$S_{j,y,c} = \frac{N_{j,y,c}}{N_{j,y}}$$

Mode shares are expressed as percentages for interpretability. These shares describe the composition of travel behaviour within each journey type and Census year, allowing direct comparison of distributional structure across 2018 and 2023.

By working with proportions rather than raw counts, differences in overall population size between education and work groups do not distort comparisons of behavioural composition.

3.2.2 Percentage-Point Change Between 2018 and 2023

To quantify temporal change within each journey type, percentage-point differences in mode shares are calculated between the two Census waves. Percentage-point change is preferred over relative percentage change because mode shares represent proportions of a total distribution. The percentage-point metric therefore directly captures structural shifts in composition.

For each journey type j and travel category c , the change between 2018 and 2023 is defined as:

$$\Delta_{j,c} = S_{j,2023,c} - S_{j,2018,c}$$

A positive value of $\Delta_{j,c}$ indicates that the share of category c increased between 2018 and 2023, while a negative value indicates a decline.

This measure allows identification of:

- growth in “At Home” participation,
- increases or decreases in private vehicle use,
- recovery or contraction in public transport usage,
- shifts toward or away from active transport.

Because the calculation is based on national population totals, the resulting changes represent actual distributional shifts in reported Census behaviour.

3.2.3 Difference-in-Differences: Education versus Work

While percentage-point change measures temporal shifts within each journey type, it does not directly quantify whether education and work evolved differently over time. To isolate this comparative effect, a difference-in-differences (DiD) measure is constructed at the travel-category level.

For each travel category c , the DiD estimator is defined as:

$$\text{DiD}_c = (S_{\text{Education},2023,c} - S_{\text{Education},2018,c}) - (S_{\text{Work},2023,c} - S_{\text{Work},2018,c})$$

Equivalently,

$$\text{DiD}_c = \Delta_{\text{Education},c} - \Delta_{\text{Work},c}$$

This statistic measures whether the change observed in education travel differs systematically from the change observed in work travel for the same category.

Interpretation:

- $\text{DiD}_c > 0$: Education experienced a larger increase (or smaller decline) than work in category c .
- $\text{DiD}_c < 0$: Work experienced a larger increase (or smaller decline) than education.
- $\text{DiD}_c = 0$: Both journey types changed by the same magnitude.

Because the analysis is based on full-population Census counts, these estimates describe structural population-level differences rather than sample-based approximations. Inferential testing of distributional differences is introduced in the following section.

3.2.4 Analytical Rationale

The combined use of national mode shares, percentage-point change, and difference-in-differences provides a structured decomposition of behavioural change:

- Mode shares describe composition at a fixed point in time.
- Percentage-point differences measure within-journey temporal shifts.
- Difference-in-differences isolates comparative evolution between education and work.

Together, these measures ensure that comparisons reflect **structural redistribution across transport categories**, rather than changes driven by overall population growth or variation in total journey counts.

3.3 Statistical Inference: Tests of Homogeneity and Distributional Change

While the descriptive measures in Section 3.2 quantify changes in travel-mode composition, formal statistical testing is required to determine whether observed differences represent meaningful distributional divergence. To evaluate homogeneity of categorical travel-mode distributions across time and journey types, Pearson’s Chi-square test is employed.

The Chi-square test of homogeneity is appropriate when comparing whether two or more independent populations share the same categorical distribution. In this study, travel categories are mutually exclusive and collectively exhaustive, and Census counts provide observed frequencies for each category.

3.3.1 Statistical Framework

To compare travel-mode distributions, contingency tables are constructed where rows represent comparison groups and columns represent travel categories. The comparison groups vary depending on the hypothesis being tested (e.g., 2018 vs 2023 within Education, or Education vs Work within 2018).

Within this framework, let

$$O_{g,c}$$

denote the observed Census count for group g and travel category c , where group g may represent:

- a Census year within a fixed journey type, or
- a journey type within a fixed Census year.

The Chi-square test evaluates whether deviations between observed and expected counts exceed what would be anticipated under the assumption of identical distributions across groups.

Under the null hypothesis of homogeneity, the expected count in each cell of the contingency table is given by:

$$E_{g,c} = \frac{R_g C_c}{N}$$

where:

- R_g is the total count for group g ,
- C_c is the total count for category c ,
- N is the grand total across all groups and categories.

The Pearson Chi-square test statistic is then defined as:

$$\chi^2 = \sum_g \sum_c \frac{(O_{g,c} - E_{g,c})^2}{E_{g,c}}$$

Under the null hypothesis, and subject to standard regularity conditions, this statistic follows a Chi-square distribution with

$$df = (G - 1)(C - 1)$$

degrees of freedom, where G is the number of groups and C is the number of travel categories.

3.3.2 Assumptions and Adequacy of the Chi-square Test

Before applying the test, it is necessary to verify that its underlying assumptions are appropriate for the structure of the aggregated Census data.

First, travel categories are mutually exclusive and collectively exhaustive within each contingency table. Each Census response contributes to exactly one main travel category, and the resulting counts therefore form a valid categorical frequency distribution within each group.

Second, the analysis is conducted using aggregated count vectors defined by group (e.g., Census year within a fixed journey type, or journey type within a fixed Census year). For each group, the vector of category counts can be treated as arising from a multinomial allocation across the five travel categories, conditional on the group total. The groups being compared represent distinct population partitions defined by Census classification. The Chi-square test therefore evaluates whether these group-specific categorical distributions differ from one another.

Third, the validity of the Chi-square approximation requires that expected cell counts are not excessively small. Given that national-level Census totals are used and the number of travel categories

is limited to five harmonised groups, expected frequencies in all cells are substantial. There is no reason to doubt the adequacy of the asymptotic approximation in this context.

Although the Census represents a full-population enumeration rather than a sample survey, the Chi-square framework remains appropriate as a distributional comparison tool. In this setting, the test quantifies the degree to which observed travel-mode distributions differ across defined population groups.

3.3.3 Time-Based Homogeneity Within Journey Types

The first set of inferential comparisons evaluates whether travel-mode distributions changed between 2018 and 2023 within each journey type.

3.3.3.1 Education: 2018 vs 2023 To determine whether the composition of education-related travel changed over time, the distribution of travel categories in 2018 is compared to that of 2023.

Let

$$\mathbf{p}_{\text{Education},y} = (S_{\text{Education},y,1}, \dots, S_{\text{Education},y,C})$$

denote the vector of travel-mode proportions for Education in Census year y .

The hypotheses are:

$$H_0 : \mathbf{p}_{\text{Education},2018} = \mathbf{p}_{\text{Education},2023}$$

$$H_A : \mathbf{p}_{\text{Education},2018} \neq \mathbf{p}_{\text{Education},2023}$$

This test evaluates whether education travel-mode composition remained stable across Census waves.

3.3.3.2 Work: 2018 vs 2023 An analogous comparison is conducted for work journeys.

Let

$$\mathbf{p}_{\text{Work},y}$$

denote the travel-mode proportion vector for Work in year y .

The hypotheses are:

$$H_0 : \mathbf{p}_{\text{Work},2018} = \mathbf{p}_{\text{Work},2023}$$

$$H_A : \mathbf{p}_{\text{Work},2018} \neq \mathbf{p}_{\text{Work},2023}$$

This assesses whether work-related travel patterns shifted between Census waves.

3.3.4 Cross-Journey Comparisons Within Census Years

The second set of tests compares education and work travel-mode distributions within a fixed Census year.

3.3.4.1 Education vs Work in 2018 Within 2018, the null hypothesis of homogeneity across journey types is:

$$H_0 : \mathbf{p}_{\text{Education},2018} = \mathbf{p}_{\text{Work},2018}$$

$$H_A : \mathbf{p}_{\text{Education},2018} \neq \mathbf{p}_{\text{Work},2018}$$

This evaluates whether students and workers exhibited identical travel-mode distributions in 2018.

3.3.4.2 Education vs Work in 2023 Similarly, for 2023:

$$H_0 : \mathbf{p}_{\text{Education},2023} = \mathbf{p}_{\text{Work},2023}$$

$$H_A : \mathbf{p}_{\text{Education},2023} \neq \mathbf{p}_{\text{Work},2023}$$

This determines whether cross-journey differences persisted or emerged in 2023.

3.3.5 Differential Change Between Journey Types

Beyond testing homogeneity at fixed points in time, it is important to assess whether the *magnitude of change* between 2018 and 2023 differs between education and work journeys.

Conceptually, this corresponds to testing for an interaction between journey type and Census year in determining travel-mode distribution.

The null hypothesis states that the distributional change over time is identical for both journey types:

$$H_0 : (\mathbf{p}_{\text{Education},2023} - \mathbf{p}_{\text{Education},2018}) = (\mathbf{p}_{\text{Work},2023} - \mathbf{p}_{\text{Work},2018})$$

The alternative hypothesis is:

$$H_A : (\mathbf{p}_{\text{Education},2023} - \mathbf{p}_{\text{Education},2018}) \neq (\mathbf{p}_{\text{Work},2023} - \mathbf{p}_{\text{Work},2018})$$

This formulation formally tests whether the structural evolution of travel behaviour differs between education and work.

3.3.6 Effect Size Assessment: Cramér's V

While the Chi-square test evaluates whether distributional differences are statistically detectable, it does not quantify the magnitude of those differences. Given the use of full-population national Census counts, very small deviations from homogeneity may yield statistically significant results. To assess the practical relevance of observed differences, effect size is therefore evaluated using Cramér's V.

Cramér's V is defined as:

$$V = \sqrt{\frac{\chi^2}{N(k-1)}}$$

where:

- χ^2 is the Pearson Chi-square statistic,
- N is the total number of observations in the contingency table,
- $k = \min(G, C)$,
- G is the number of groups,
- C is the number of travel categories.

Cramér's V ranges between 0 and 1, with larger values indicating stronger departure from homogeneity.

Although interpretation thresholds depend on context and table dimensions, conventional benchmarks are often used for guidance:

- $V \approx 0.10$: small association
- $V \approx 0.30$: moderate association
- $V \approx 0.50$: large association

These thresholds are interpreted cautiously, given the aggregated and population-level nature of the data.

Reporting Cramér's V alongside p-values ensures that conclusions regarding distributional change are informed not only by statistical significance but also by substantive magnitude.

3.4 Multinomial Modelling of Time-Journey Interaction Effects

While Chi-square tests assess distributional homogeneity across pairs of groups, they do not directly model the joint influence of Census year and journey type on travel-mode probabilities. To formally evaluate whether temporal shifts in travel-mode composition differ between Education and Work journeys, a multinomial logistic regression model with interaction terms is employed.

This model provides a parametric framework for analysing categorical outcomes with more than two unordered categories. In this context, the outcome variable is travel category, which consists of five mutually exclusive categories:

- Private Transport
- Public Transport
- Active Transport
- At Home
- Other Transport

The modelling objective is to determine whether the change in travel-mode probabilities between 2018 and 2023 depends on journey type.

3.4.1 Conceptual Framework

The multinomial modelling framework is introduced to evaluate whether the evolution of travel-mode composition over time differs systematically between Education and Work journeys. While earlier sections tested homogeneity using contingency tables, those approaches evaluate distributions pairwise. The multinomial model instead provides a unified probability-based representation of travel behaviour, allowing time effects, journey-type effects, and their interaction to be assessed simultaneously.

In this context, travel mode is treated as a categorical outcome with multiple unordered categories. For each combination of Census year and journey type, individuals are distributed across a fixed set of travel categories. The modelling objective is to describe how the probability of each travel category varies as a function of Census year and journey type, and to determine whether the temporal shift from 2018 to 2023 differs between the two journey contexts.

Formally, let

$$Y_i \in \{1, 2, \dots, C\}$$

denote the travel category for observation i , where $C = 5$ represents the five harmonised travel categories.

Let:

- $y \in \{2018, 2023\}$ denote Census year,
- $j \in \{\text{Education}, \text{Work}\}$ denote journey type.

The probability of travel category c conditional on Census year and journey type is defined as:

$$\pi_{j,y,c} = P(Y = c \mid j, y)$$

The central inferential question is whether the change in these probabilities between 2018 and 2023 differs across journey types. In other words, the analysis tests whether the effect of time on travel-mode composition depends on whether the journey is for education or for work. Within a regression framework, this corresponds to testing for the presence of a statistical interaction between Census year and journey type.

3.4.2 Multinomial Logistic Model Specification

Having defined the conceptual objective of modelling travel-mode probabilities as a function of Census year and journey type, we now formalise the regression structure used to estimate these effects.

Because travel mode consists of five unordered categories, a multinomial logistic regression model is employed. This model expresses the log-odds of selecting each non-reference travel category relative to a baseline category as a linear function of explanatory variables.

One category must be chosen as a reference. Let category C serve as the baseline. All other categories are modelled relative to this reference.

For each category $c \neq C$, the log-odds of choosing category c over the reference category is modelled as:

$$\log \left(\frac{\pi_{j,y,c}}{\pi_{j,y,C}} \right) = \beta_{0c} + \beta_{1c} \text{Year}_y + \beta_{2c} \text{Journey}_j + \beta_{3c} (\text{Year}_y \times \text{Journey}_j)$$

where:

- $\pi_{j,y,c}$ is the probability of travel category c
- β_{0c} is the baseline log-odds for category c
- β_{1c} captures the main effect of Census year
- β_{2c} captures the main effect of journey type
- β_{3c} captures the interaction between year and journey type

The interaction term allows the effect of time (2018 vs 2023) to differ between Education and Work journeys.

In the absence of interaction ($\beta_{3c} = 0$), the time-based change in travel-mode probabilities is identical across journey types. When interaction terms are non-zero, temporal shifts vary systematically between Education and Work.

This specification provides a unified parametric framework for modelling structural change across multiple travel categories simultaneously.

3.4.3 Hypothesis Testing: Interaction Between Year and Journey Type

The primary inferential objective of the multinomial model is to determine whether time-based shifts in travel-mode composition differ between Education and Work journeys.

Conceptually, this corresponds to testing whether the effect of Census year depends on journey type - that is, whether an interaction is present.

Formally, the null hypothesis states that the interaction coefficients are jointly zero across all non-reference categories:

$$H_0 : \beta_{3c} = 0 \quad \text{for all } c$$

Under this null hypothesis, the temporal shift from 2018 to 2023 is identical for Education and Work journeys.

The alternative hypothesis is:

$$H_A : \exists c \text{ such that } \beta_{3c} \neq 0$$

indicating that at least one travel category exhibits a differential time-based change between journey types.

To evaluate this hypothesis, a likelihood ratio test is conducted. Two nested models are compared:

1. A reduced model containing only main effects (Year and Journey type),
2. A full model containing both main effects and interaction terms.

Let:

- L_{reduced} denote the likelihood of the model without interaction,
- L_{full} denote the likelihood of the model with interaction.

The likelihood ratio statistic is defined as:

$$\Lambda = -2 (\log L_{\text{reduced}} - \log L_{\text{full}})$$

Under standard regularity conditions, this statistic follows a Chi-square distribution with degrees of freedom equal to the number of interaction parameters removed.

Rejection of H_0 implies that the evolution of travel-mode probabilities between 2018 and 2023 differs between Education and Work journeys.

This modelling approach provides a direct and unified test of the central research question regarding differential structural change.

3.4.4 Model Assumptions and Validity

The validity of the multinomial logit model relies on several assumptions:

3.4.4.1 Multinomial Count Formulation and Group Structure The multinomial model is fitted to aggregated counts defined by journey type and Census year across the five harmonised travel categories. For each combination (j, y) , the observed count vector $(N_{j,y,1}, \dots, N_{j,y,C})$ is treated as a multinomial outcome conditional on the total $N_{j,y}$.

This formulation is consistent with the mutually exclusive and collectively exhaustive nature of the Census-reported main travel mode variable.

Inference is framed around differences in categorical distributions between group-defined populations (journey type and Census year), rather than independence of individual unit-record observations. Because the analysis operates on aggregated count vectors rather than microdata, the relevant assumption is the multinomial allocation of counts within each group.

Given the structure of the Census tabulations and the use of national-level totals, this modelling framework is appropriate.

3.4.4.2 Multinomial Sampling Structure Within each combination of year and journey type, the counts across travel categories sum to a fixed total. This is consistent with a multinomial data-generating process.

Given that national-level counts are used, expected cell frequencies are substantial, and asymptotic approximations are reliable.

3.4.4.3 Independence of Irrelevant Alternatives (IIA) The multinomial logit model assumes that the relative odds between any two categories are independent of the presence of other categories.

In this context, travel categories are broad structural groupings rather than highly substitutable alternatives (e.g., “Private”, “Public”, “Active”, etc.). While no model perfectly satisfies IIA in behavioural contexts, there is no empirical indication that category aggregation induces violations severe enough to invalidate inference.

Given the categorical aggregation strategy and the policy-oriented objective of the analysis, the multinomial logit framework is considered appropriate.

3.4.4.4 Absence of Perfect Separation Given large Census counts across all categories and groups, there is no evidence of zero-probability categories that would induce separation issues.

3.4.5 Effect Size Interpretation in the Multinomial Framework

While likelihood ratio tests determine whether the interaction between Census year and journey type is statistically significant, statistical significance alone does not convey the magnitude of the effect. Given the large national Census counts used in this study, even relatively small shifts in travel-mode probabilities may yield highly significant test statistics. It is therefore essential to complement hypothesis testing with measures that reflect substantive magnitude.

In a multinomial logistic framework, effect size is interpreted through the estimated regression coefficients and their transformation into odds ratios. Each coefficient represents the change in the log-odds of selecting travel category c relative to the reference category, associated with a unit change in the predictor.

For category $c \neq C$, the interaction coefficient β_{3c} quantifies how the time-based change in log-odds differs between Education and Work journeys.

Exponentiating the coefficient yields the odds ratio:

$$\text{OR}_{3c} = \exp(\beta_{3c})$$

An odds ratio interpretation follows:

- $\text{OR}_{3c} = 1$: No differential time effect between journey types
- $\text{OR}_{3c} > 1$: The increase in odds over time is larger for Education relative to Work (or vice versa depending on coding)
- $\text{OR}_{3c} < 1$: The increase in odds over time is smaller for Education relative to Work

Unlike the Chi-square statistic, odds ratios are scale-independent and allow interpretation of direction and relative magnitude.

However, odds ratios reflect changes in relative odds rather than changes in actual probabilities. Because policy interpretation is often more intuitive in probability space, predicted probabilities are additionally computed.

3.4.6 Predicted Probability Formulation and Interpretation

Regression coefficients in a multinomial logit model are expressed in log-odds space, which can be difficult to interpret substantively. To translate model estimates into meaningful behavioural interpretation, predicted probabilities are derived for each combination of Census year and journey type.

The multinomial logit model specifies category probabilities as:

$$\pi_{j,y,c} = \frac{\exp(\eta_{j,y,c})}{\sum_{k=1}^C \exp(\eta_{j,y,k})}$$

where:

$$\eta_{j,y,c} = \beta_{0c} + \beta_{1c} \text{Year}_y + \beta_{2c} \text{Journey}_j + \beta_{3c} (\text{Year}_y \times \text{Journey}_j)$$

and the reference category satisfies $\eta_{j,y,C} = 0$.

This formulation ensures that:

$$\sum_{c=1}^C \pi_{j,y,c} = 1$$

for every combination of year and journey type.

Predicted probabilities provide a direct estimate of:

$$P(Y = c \mid j, y)$$

for each travel category.

These probabilities allow:

- Direct comparison of travel-mode shares between 2018 and 2023
- Direct comparison between Education and Work
- Quantification of the magnitude of time-based shifts
- Clear visual presentation of interaction effects

Importantly, the presence of a statistically significant interaction implies that:

$$\pi_{\text{Education},2023,c} - \pi_{\text{Education},2018,c} \neq \pi_{\text{Work},2023,c} - \pi_{\text{Work},2018,c}$$

for at least one category c .

Thus, predicted probabilities translate statistical interaction into interpretable distributional change.

3.4.7 Rationale for Model Choice

The multinomial interaction model offers several methodological advantages over pairwise Chi-square testing.

First, it evaluates interaction effects directly within a single unified regression framework, rather than relying on multiple contingency-table comparisons. This allows the time effect, journey-type effect, and their interaction to be estimated simultaneously.

Second, the model provides category-specific parameter estimates, enabling identification of which travel categories contribute most strongly to differential structural change.

Third, likelihood-based model comparison provides a formal and coherent inferential basis for testing nested hypotheses regarding interaction effects.

Fourth, the regression framework facilitates effect size interpretation through odds ratios, allowing magnitude and direction of change to be quantified beyond mere statistical significance.

Finally, the derivation of predicted probabilities enables interpretation in probability space rather than log-odds space. This is particularly important in a policy context, where percentage-point changes in travel-mode shares are more interpretable than relative odds. The model therefore bridges statistical inference and substantive interpretation.

Taken together, the multinomial interaction framework provides a comprehensive and interpretable approach to assessing whether time-based shifts in travel-mode distributions differ between Education and Work journeys.

There is no evidence suggesting that the underlying model assumptions are violated in a manner that would preclude its use in this setting.

3.5 Extensions to Regions and Sub-Populations

While Sections 3.2-3.4 focus on national-level travel-mode composition and inferential comparison, structural change in travel behaviour may not be spatially uniform. Urban form, settlement density, transport infrastructure, and institutional arrangements vary considerably across New Zealand. To assess spatial heterogeneity, the analytical framework defined previously is extended to the regional and city level.

This section applies the previously defined measures of mode share, percentage-point change, and difference-in-differences at finer geographic resolution. No new inferential model is introduced; rather, the national framework is implemented within regional partitions of the data.

3.5.1 Region-Level Travel Mode Shares and Percentage-Point Change

For each region r , journey type j , Census year y , and travel category c , regional mode shares are computed using the same formulation defined in Section 3.2.1, but restricting aggregation to region r .

Thus, regional shares take the form:

$$S_{r,j,y,c}$$

where totals are summed over all SA2 units belonging to region r .

Temporal change within each region and journey type is then measured using the percentage-point change formulation introduced in Section 3.2.2:

$$\Delta_{r,j,c} = S_{r,j,2023,c} - S_{r,j,2018,c}$$

This produces a regional map of structural change, allowing identification of regions experiencing larger or smaller shifts in specific travel categories.

3.5.2 Regional Difference-in-Differences

To evaluate whether temporal shifts differ between Education and Work within a specific region, the difference-in-differences framework from Section 3.2.3 is applied at the regional level.

For each region r and category c :

$$DiD_{r,c} = \Delta_{r,Education,c} - \Delta_{r,Work,c}$$

This statistic quantifies whether the magnitude of change between 2018 and 2023 differs between journey types within region r .

Positive values indicate that Education experienced a larger increase (or smaller decline) than Work in that region, while negative values indicate the opposite.

These quantities remain descriptive and are not subjected to separate inferential testing.

3.5.3 Urban and Non-Urban Group Comparisons

Because settlement structure may systematically influence travel behaviour, regions are grouped into Urban and Non-Urban classifications. To summarise behaviour at the group level, weighted averages of regional shares are computed.

Let $N_{r,j,y}$ denote the total number of journeys in region r for journey type j and Census year y .

Group-level mode shares are defined as weighted averages:

$$S_{G,j,y,c} = \frac{\sum_{r \in G} N_{r,j,y} S_{r,j,y,c}}{\sum_{r \in G} N_{r,j,y}}$$

where $G \in \{\text{Urban}, \text{Non-Urban}\}$.

This weighting ensures that regions contribute proportionally according to their population size, preventing small regions from exerting disproportionate influence.

Percentage-point changes and difference-in-differences at the group level are then computed using the same definitions provided in Sections 3.2.2 and 3.2.3.

3.5.4 City-Level Comparison

To provide finer-grained insight within selected metropolitan and regional centres, four representative territorial authorities were analysed separately: Wellington City and Christchurch City (metropolitan centres), and Gisborne District and Invercargill City (non-metropolitan centres). These locations were selected to contrast large urban labour-market hubs with smaller regional districts.

For each city:

- Mode shares are computed as in Section 3.2.1.
- Percentage-point changes are computed as in Section 3.2.2.
- Difference-in-differences is computed as in Section 3.2.3.

This extension enables comparison of whether modal redistribution differs between major metropolitan centres and smaller regional districts.

3.5.5 Age-Group Stratified Analysis

Travel behaviour may also vary systematically across life stages. To assess demographic heterogeneity, the analytical framework is extended to age-group stratified comparisons. The analysis focuses on broad Census age categories that are consistently defined across both Journey to Education and Journey to Work datasets. Because the Work dataset is restricted to individuals aged 15 years and over, difference-in-differences comparisons exclude the Under-15 cohort. For each age group a , journey type j , Census year y , and travel category c , age-specific mode shares are computed using the formulation defined in Section 3.2.1, restricting aggregation to age group (a):

$$S_{a,j,y,c} = \frac{N_{a,j,y,c}}{N_{a,j,y}}$$

where:

$$N_{a,j,y} = \sum_{c=1}^C N_{a,j,y,c}.$$

Temporal change within each age group and journey type is measured using the percentage-point definition introduced in Section 3.2.2:

$$\Delta_{a,j,c} = S_{a,j,2023,c} - S_{a,j,2018,c}.$$

To evaluate whether temporal shifts differ between Education and Work within a given age group, the difference-in-differences framework defined in Section 3.2.3 is applied at the age-group level:

$$DiD_{a,c} = \Delta_{a,Education,c} - \Delta_{a,Work,c}.$$

These age-specific quantities are descriptive extensions of the national framework and allow assessment of whether modal restructuring is uniformly distributed across cohorts or concentrated within specific age groups.

3.5.6 Analytical Positioning

The regional, urban-non-urban, city-level, and age-group extensions serve to assess spatial and demographic heterogeneity in modal redistribution patterns. These analyses complement the national-level inferential results presented earlier but do not introduce additional model-based inference.

Together, they provide a structured decomposition of how Education and Work travel evolved across geographic contexts and life-stage cohorts in the post-pandemic period.

3.5.7 Software and Reproducibility

All statistical analyses were conducted in R (version 4.4.3) within RStudio. Data manipulation was performed using `dplyr` and `tidyr`, and multinomial logistic regression models were estimated using the `nnet` package. Tables were generated programmatically within an R Markdown workflow to ensure full reproducibility.

The analytical workflow was implemented through a fully scripted and deterministic R Markdown pipeline, ensuring that all reported tables, statistical tests, and visualisations are reproducible directly from the cleaned dataset. No manual post-processing was applied to statistical outputs.

Interactive visualisations and dashboard summaries were additionally developed using Microsoft Power BI Desktop. Custom DAX measures were implemented to compute dynamic mode shares, percentage-point changes, and Education-Work gaps for exploratory analysis and presentation purposes. The Power BI project file (.pbix), containing all report pages, calculated measures, and visual configurations, is included in the project repository.

The complete reproducible analysis pipeline—including data preparation scripts, statistical modelling code, table and figure generation files, and the Power BI dashboard file, which is publicly available in a version-controlled repository at: <https://github.com/tanveer09/move-nz>

The repository contains the full R Markdown workflow and supporting documentation required to replicate all results reported in this study.

4 Results

This section presents the empirical findings corresponding to the analytical framework defined in Sections 3.2-3.5. Results are structured to mirror the methodological sequence, beginning with national-level travel-mode composition and percentage-point changes, followed by formal statistical inference, multinomial interaction modelling, regional extensions, and subgroup driver analysis.

All reported quantities are derived from aggregated Census population counts for 2018 and 2023. Descriptive results are presented using mode shares and percentage-point changes, while inferential findings include Chi-square tests of homogeneity, effect size measures, and likelihood-based model comparisons. Predicted probabilities from the multinomial model are used to facilitate interpretation in percentage-point space.

Figures reproduced from the Power BI analytical dashboard are included to visualise structural shifts, while corresponding statistical outputs and model summaries are presented in tables derived from the R analysis pipeline. Together, these results provide a comprehensive evaluation of whether and how travel behaviour to education and work evolved between Census waves.

4.1 National Travel-Mode Composition and Percentage-Point Change

This subsection presents national travel-mode distributions for Journey to Education and Journey to Work in the 2018 and 2023 Censuses. Mode shares are computed from aggregated national counts and expressed as percentages to enable direct comparison of distributional composition across journey types and Census waves.

In 2018, work-related travel was dominated by Private Transport, accounting for the majority of journeys, followed by Public Transport and Active Transport. Education journeys exhibited a more diversified distribution, with relatively higher shares in Active and Public Transport compared to Work, and a smaller reliance on Private Transport.

By 2023, both journey types exhibit a marked structural shift. The most notable change is the substantial increase in the “At home” category for both Education and Work. This increase reflects a redistribution away from traditional commuting modes. Correspondingly, Public Transport shares decline relative to 2018 levels, while Private Transport and Active Transport display more moderate adjustments.

The comparison of national-level mode shares reveals that although both journey types moved in similar directions between Census waves, the magnitude of change differs across categories. Education maintains a comparatively higher reliance on Active and Public modes than Work in both years, whereas Work remains more concentrated in Private Transport.

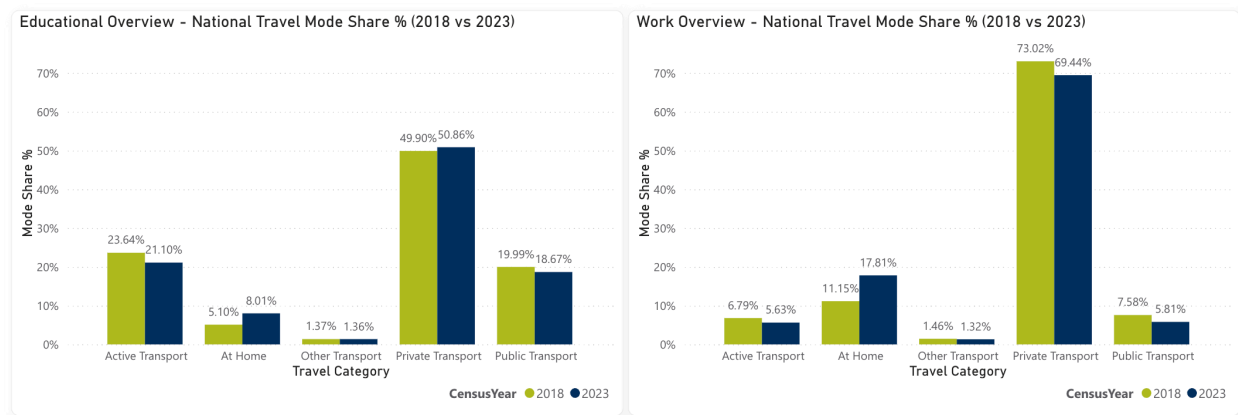


Figure 1: National Travel Mode Share Comparison: Education vs Work (2018 vs 2023)

Table 2: National travel mode share (%) in 2018 and 2023, shown separately for Education and Work journeys

Travel Category	Education 2018 (%)	Education 2023 (%)	Work 2018 (%)	Work 2023 (%)
Active Transport	23.64	21.10	6.79	5.63
At Home	5.10	8.01	11.15	17.81
Other Transport	1.37	1.36	1.46	1.32
Private Transport	49.90	50.86	73.02	69.44
Public Transport	19.99	18.67	7.58	5.81

4.1.1 Percentage-Point Change (2018-2023)

To quantify structural change over time, percentage-point differences between 2018 and 2023 are examined for each travel category within each journey type.

The most pronounced change for both Education and Work is the increase in the “At Home” category. This increase represents a structural redistribution in travel behaviour rather than a marginal fluctuation. Public Transport exhibits a clear decline across both journey types, indicating contraction relative to pre-2018 levels.

Private Transport displays contrasting behaviour across journey types. While it increases modestly for Education, it declines substantially for Work, reflecting a meaningful shift away from traditional commuting patterns. Active Transport demonstrates relatively modest percentage-point variation compared to the larger shifts observed in home-based activity.

Although several directional patterns of change are similar across Education and Work, the magnitude and direction of percentage-point shifts differ across categories. This motivates the comparative analysis presented next.

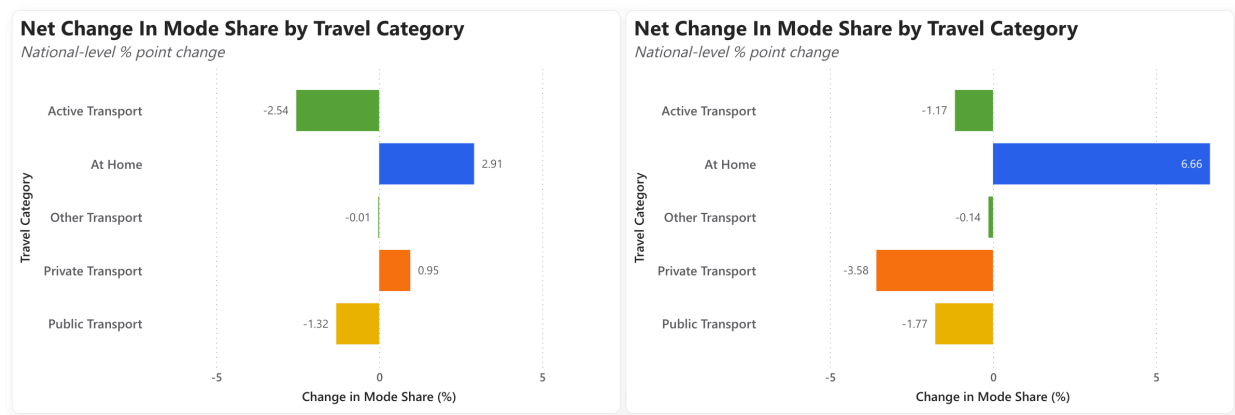


Figure 2: Percentage-Point Change: Education vs Work (2018-2023)

Table 3: Percentage-point change in national travel mode share between 2018 and 2023, by journey type

Travel Category	Education (pp change 2018–2023)	Work (pp change 2018–2023)
Active Transport	-2.54	-1.16
At Home	2.91	6.66
Other Transport	-0.01	-0.14
Private Transport	0.96	-3.58
Public Transport	-1.32	-1.77

4.1.2 Difference-in-Differences: Education vs Work

To determine whether Education and Work evolved differently over time, difference-in-differences estimates are examined at the category level.

The DiD results indicate that although both journey types experienced increases in “At Home,” the magnitude of change differs substantially, with Work exhibiting a larger increase than Education. Reductions in Public Transport occur for both journey types, though the divergence between them is comparatively modest. Private Transport displays the strongest differential effect, increasing slightly for Education but declining markedly for Work.

Categories with DiD values near zero suggest broadly parallel structural shifts across journey types, whereas larger positive or negative DiD values indicate meaningful divergence. In particular, the contrasting behaviour in Private Transport and the stronger rise in “At Home” for Work highlight that the structural reallocation of travel modes is not identical between Education and Work contexts.

This descriptive divergence sets the stage for the formal inferential tests presented in Section 4.2.

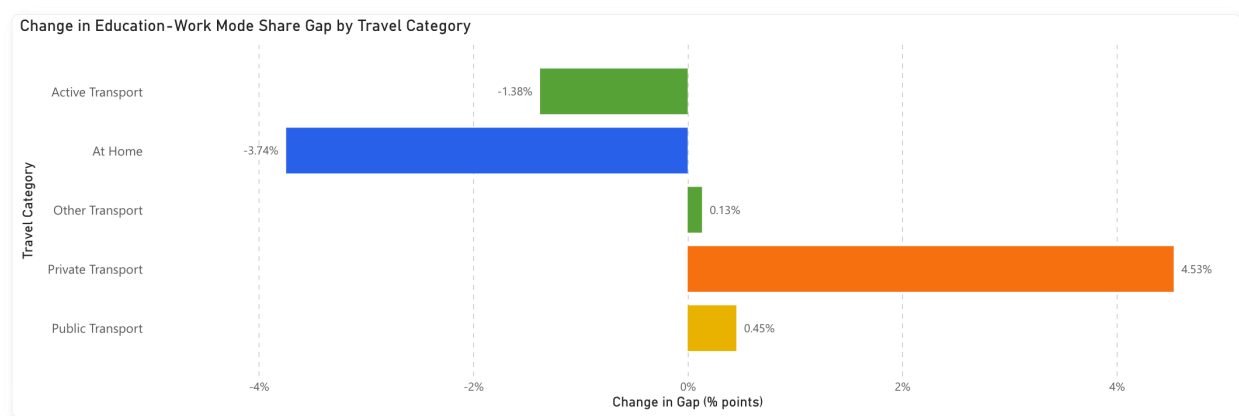


Figure 3: Difference-in-Differences: Education vs Work (2018-2023)

Table 4: Descriptive difference-in-differences: Education vs Work percentage-point shifts (2018–2023), by travel category

Travel Category	Education (pp change 2018–2023)	Work (pp change 2018–2023)	DiD (Education – Work, pp)
Active Transport	-2.54	-1.16	-1.38
At Home	2.91	6.66	-3.75
Other Transport	-0.01	-0.14	0.13
Private Transport	0.96	-3.58	4.54
Public Transport	-1.32	-1.77	0.45

4.2 Statistical Inference: Distributional Comparisons Across Years and Journey Types

Section 4.1 established descriptive differences in travel-mode composition across Census years and journey types. This section formalises those observations using Pearson’s chi-square tests to evalu-

ate whether observed differences represent statistically detectable distributional divergence. Given the extremely large national Census counts, statistical significance is interpreted alongside effect size (Cramér’s V) to distinguish detectability from substantive magnitude.

4.2.1 Time-Based Change Within Education (2018 vs 2023)

Is there statistically significant evidence that the distribution of travel modes for Education changed between 2018 and 2023?

The Pearson chi-square test showed a statistically significant association between Census year and travel category within Education journeys ($\chi^2(4) = 13,501, p < 0.001$). Since $p < 0.05$, we reject the null hypothesis that the distribution of Education travel modes was identical in 2018 and 2023. This indicates that education-related travel behaviour underwent statistically detectable structural redistribution over the five-year period.

Table 5: Pearson chi-square test for time-based changes in travel mode distribution (Education)

Journey Type	Statistic	Value	df	p_value
Education	Chi-square	13501	4	<1e-16

4.2.2 Time-Based Change Within Work (2018 vs 2023)

Is there statistically significant evidence that the distribution of travel modes for Work changed between 2018 and 2023?

The Pearson chi-square test showed a statistically significant association between Census year and travel category within Work journeys ($\chi^2(4) = 66,534, p < 0.001$). Since $p < 0.05$, we reject the null hypothesis that the distribution of Work travel modes was identical in 2018 and 2023. This indicates that work-related travel patterns experienced substantial structural change between Census waves.

Table 6: Pearson chi-square test for time-based changes in travel mode distribution (Work)

Journey Type	Statistic	Value	df	p_value
Work	Chi-square	66534	4	<1e-16

4.2.3 Cross-Journey Comparison in 2018

Is there statistically significant evidence that the distribution of travel modes differed between Education and Work in 2018?

The Pearson chi-square test showed a statistically significant association between journey type and travel category in 2018 ($\chi^2(4) = 517,341, p < 0.001$). Since $p < 0.05$, we reject the null hypothesis that Education and Work shared the same travel-mode distribution in 2018. This confirms that baseline modal composition differed structurally between students and workers prior to the pandemic period.

Table 7: Pearson’s chi-squared test comparing Education and Work travel mode distributions in 2018

Census Year	Test	Chi_square	df	p_value
2018	Pearson’s chi-squared test	517341	4	<1e-16

4.2.4 Cross-Journey Comparison in 2023

Is there statistically significant evidence that the distribution of travel modes differed between Education and Work in 2023?

The Pearson chi-square test showed a statistically significant association between journey type and travel category in 2023 ($\chi^2(4) = 572,042, p < 0.001$). Since $p < 0.05$, we reject the null hypothesis that Education and Work shared the same travel-mode distribution in 2023. This indicates that journey-purpose differentiation in travel behaviour persisted in the post-pandemic period.

Table 8: Pearson’s chi-squared test comparing Education and Work travel mode distributions in 2023

Census Year	Test	Chi_square	df	p_value
2023	Pearson’s chi-squared test	572042	4	<1e-16

4.2.5 Effect Size for Cross-Journey Differences

Because Census counts are extremely large, statistical significance alone does not convey substantive magnitude. Effect size is therefore assessed using Cramér’s V.

The estimated values are 0.3261 for 2018 and 0.3339 for 2023, indicating a moderate-to-strong association between journey type and travel category in both Census years. The stability of effect size across waves suggests that cross-journey structural differentiation remains substantively meaningful rather than merely statistically detectable.

Table 9: Cramér’s V effect size for the association between journey type and travel category (Education vs Work), by census year

Census Year	Cramér’s V
2018	0.3261
2023	0.3339

4.2.6 Differential Change Between Journey Types (2018-2023)

Is there statistically significant evidence that the change in travel-mode distributions between 2018 and 2023 differed between Education and Work journeys?

The Pearson chi-square test showed a statistically significant association between journey type and category-wise change profiles over time ($\chi^2(4) = 91,461, p < 0.001$). Since $p < 0.05$, we reject the null hypothesis that Education and Work experienced identical temporal redistribution across travel categories. This indicates that the magnitude and structure of modal change between 2018 and 2023 differed systematically between journey purposes.

Table 10: Pearson’s chi-squared test comparing the magnitude of category-wise change profiles between Education and Work (2018–2023)

Test	Chi_square	df	p_value
Pearson’s chi-squared test	91461	4	<1e-16

Synthesis of Inferential Findings

Collectively, the chi-square analyses establish three consistent findings. First, both Education and Work travel-mode distributions changed significantly between 2018 and 2023. Second, Education and Work differed significantly within each Census year, with moderate-to-strong effect sizes confirming substantive journey-purpose differentiation. Third, the structure of temporal redistribution differed between journey types, indicating that post-pandemic travel adjustments were not uniform across journey purposes. Together, these results provide strong statistical evidence that travel behaviour in New Zealand underwent structurally differentiated change between 2018 and 2023.

4.3 Multinomial Model: Differential Time-Based Change Between Education and Work

To complement the contingency-table analyses presented in Section 4.2, a multinomial logistic regression model with a $Year \times JourneyType$ interaction was estimated, as described in Section 3.4.

While Section 4.2 established statistically significant distributional differences using contingency-table methods, this section provides a unified regression-based assessment of whether the 2018-2023 shifts differ systematically between Education and Work journeys.

A multinomial logistic regression model was estimated with travel category as the outcome and Census year, journey type, and their interaction as predictors. The key inferential focus is the $Year \times Journey Type$ interaction, which tests whether the temporal effect of Census year differs across journey purposes.

4.3.1 Likelihood Ratio Test for $Year \times Journey Type$ Interaction

Is there statistically significant evidence, based on the multinomial interaction model, that the change in travel-mode distributions between 2018 and 2023 differed between Education and Work journeys?

A likelihood ratio test compared a baseline multinomial model containing only main effects (Census year and journey type) with a model including the $Year \times Journey Type$ interaction.

The likelihood ratio test showed a statistically significant improvement in model fit when the interaction term was included ($\chi^2(4) = 1320, p < 0.001$).

Table 11: Likelihood-ratio test for the Year $\times$ Journey Type interaction in the multinomial logistic regression model

Likelihood Ratio Statistic	df	p_value
1320	4	<1e-16

Since $p < 0.05$, we reject the null hypothesis that temporal shifts in travel-mode distribution were identical for Education and Work journeys. This indicates that changes between 2018 and 2023 differed significantly by journey type.

Table 12: Likelihood ratio tests of multinomial models

Model	Resid. df	Resid. Dev	Test	Df	LR stat.	Pr(Chi)
census_year + journey_type	4232820	20533275		NA	NA	NA
census_year * journey_type	4232816	20531955	1 vs 2	4	1320.175	0

Table 12 reports the model comparison results. The inclusion of the Year $\times$ Journey Type interaction significantly improves model fit.

4.3.2 Model-Based Percentage-Point Changes in Travel Mode Shares

While the likelihood ratio test establishes statistical significance of the interaction, it does not directly convey how travel behaviour changed. To obtain interpretable effect sizes, predicted probabilities from the fitted multinomial model were computed and converted into percentage-point changes for each journey type and travel category.

The model-based predicted shares confirm distinct temporal adjustment patterns across journey types.

For Education journeys, the predicted percentage-point changes between 2018 and 2023 are as follows:

Table 13: Model-based predicted travel mode shares (%) in 2018 and 2023 and percentage-point change (2023 – 2018): Education

Travel Category	Predicted share 2018 (%)	Predicted share 2023 (%)	PP change (2023 – 2018)
Active Transport	23.64	21.10	-2.54
At Home	5.10	8.01	2.91
Other Transport	1.37	1.37	0.00
Private Transport	49.90	50.86	0.96
Public Transport	19.99	18.67	-1.32

For Work journeys, the predicted percentage-point changes are as follows:

Table 14: Model-based predicted travel mode shares (%) in 2018 and 2023 and percentage-point change (2023 – 2018): Work

Travel Category	Predicted share 2018 (%)	Predicted share 2023 (%)	PP change (2023 – 2018)
Active Transport	6.79	5.63	-1.16
At Home	11.15	17.81	6.66
Other Transport	1.46	1.32	-0.14
Private Transport	73.02	69.44	-3.58
Public Transport	7.58	5.81	-1.77

These results demonstrate that although both journey types experienced increases in at-home travel, the magnitude of increase was substantially larger for Work. Conversely, Education exhibited a modest increase in private transport, whereas Work showed a pronounced decline.

This confirms that the interaction detected by the likelihood ratio test reflects meaningful and interpretable structural divergence in modal reallocation.

4.3.3 Interpretation and Synthesis

The multinomial modelling results reinforce and extend the contingency-table findings in Section 4.2. The significant interaction ($\chi^2(4) = 1320, p < 0.001$) confirms that the 2018-2023 shift in travel-mode distribution differs between Education and Work when all travel categories are considered jointly.

Importantly, the predicted probability contrasts reveal the substantive structure of this interaction: Work journeys experienced a substantially larger shift toward at-home participation and a notable decline in private transport, whereas Education journeys underwent more moderate adjustments.

Together, the likelihood ratio test and model-based predicted probabilities provide converging evidence that post-2018 modal reallocation was not uniform across journey purposes. The regression framework therefore confirms that the observed difference-in-differences patterns reflect statistically robust and substantively meaningful divergence between Education and Work travel behaviour.

4.4 Regional and Sub-Population Extensions

This subsection extends the national-level findings by examining whether modal redistribution between 2018 and 2023 is spatially uniform or regionally differentiated across geographic and demographic contexts. Consistent with Section 2.5, the analysis proceeds in sequential stages: (i) region-level distributional and percentage-point change, (ii) urban versus non-urban comparisons, (iii) city-level contrasts between selected metropolitan and non-metropolitan centres, and (iv) age-group differences in modal restructuring, including Education–Work difference-in-differences. All analyses are descriptive and based on aggregated Census population counts.

4.4.1 Region-Level Distributional Change

To assess whether travel-mode distributions changed within each region between the 2018 and 2023 Censuses, Pearson Chi-square tests of homogeneity were conducted separately for Journey to Education and Journey to Work.

Table 15 below reports the region-specific Chi-square statistics, degrees of freedom, and associated p-values for both journey purposes.

Table 15: Region-wise Pearson chi-square tests of homogeneity for changes in travel mode distributions between 2018 and 2023

Region	Edu chi_sq	Edu df	Edu p-value	Work chi_sq	Work df	Work p-value
Auckland Region	8436.39	4	0.00e+00	68920.84	4	0.00e+00
Canterbury Region	1050.14	4	4.86e-226	1588.73	4	0.00e+00
Wellington Region	1479.95	4	3.19e-319	11793.45	4	0.00e+00
Waikato Region	1008.46	4	5.25e-217	1291.04	4	2.92e-278
Manawatu-Whanganui Region	722.42	4	4.88e-155	654.48	4	2.50e-140
Bay of Plenty Region	634.14	4	6.33e-136	649.68	4	2.74e-139
Northland Region	412.35	4	5.95e-88	447.26	4	1.69e-95
Hawke's Bay Region	446.62	4	2.33e-95	313.16	4	1.57e-66
Otago Region	305.88	4	5.85e-65	437.28	4	2.44e-93
Gisborne Region	255.75	4	3.75e-54	63.13	4	6.38e-13
Taranaki Region	186.50	4	2.99e-39	150.70	4	1.44e-31
Southland Region	156.96	4	6.55e-33	19.47	4	6.36e-04
West Coast Region	128.23	4	9.31e-27	41.49	4	2.13e-08
Nelson Region	105.11	4	8.04e-22	109.86	4	7.78e-23
Tasman Region	42.20	4	1.52e-08	61.95	4	1.13e-12
Marlborough Region	57.33	4	1.06e-11	60.84	4	1.93e-12
Area Outside Region	9.97	4	4.09e-02	11.11	4	2.54e-02

Across all regions, the tests provide strong evidence of distributional change for both Education and Work. P-values are uniformly extremely small, indicating statistically detectable shifts in modal composition between Census waves. Given the population-level nature of Census data, statistical significance is expected; therefore, interpretation emphasises the relative magnitude of change rather than significance alone.

The magnitude of the Chi-square statistics varies substantially across regions. Major metropolitan regions (e.g., Auckland, Wellington, Canterbury) exhibit markedly larger statistics, particularly for Work, reflecting substantial redistribution in commuting patterns. In contrast, smaller regions display lower absolute values, although distributional change remains statistically evident.

Notably, in most regions the Chi-square statistics for Work exceed those for Education, indicating that modal restructuring between 2018 and 2023 was generally stronger for Work-related travel. This pattern is consistent with broader structural adjustments in commuting behaviour.

These results confirm that modal change occurred across all regions and for both journey purposes, motivating further examination of the magnitude and directional composition of regional percentage-point changes.

4.4.2 Region-Level Percentage-Point Change

To quantify the magnitude and direction of change, percentage-point (pp) differences in mode share between 2018 and 2023 were computed for each region and journey type. Full results are reported in **Table A1 (Education) and Table A2 (Work) in the Appendix**.

Across regions, at-home participation increased for both journey types, with consistently larger gains observed for Work. The strongest increases are concentrated in major metropolitan regions, including Wellington and Auckland.

Private transport declines are more pronounced for Work than for Education in metropolitan regions, whereas Education-related private transport changes are smaller and more heterogeneous.

Public transport generally declines in high-reliance metropolitan regions, particularly for Work. Active transport also decreases in most regions, with larger reductions observed in several smaller regions for Education.

Overall, modal redistribution occurred nationwide, but the magnitude of change is greater for Work and spatially concentrated in metropolitan centres.

4.4.3 Regional Difference-in-Differences

To quantify divergence between Education and Work within each region, regional difference-in-differences (DiD) measures were computed as:

$$DiD = (\Delta Education) - (\Delta Work)$$

where Δ denotes the 2018-2023 percentage-point change.

Regional DiD results are reported in **Table A7 in the Appendix**.

Positive DiD values indicate that the increase (or reduction) in mode share was larger for Education than for Work, while negative values indicate stronger change for Work.

Across regions, the largest negative DiD values occur in the **At-Home** category in major metropolitan regions, particularly Wellington and Auckland, indicating that remote participation increased substantially more for Work than for Education in these centres.

In contrast, several regions exhibit positive DiD values in **Private Transport**, reflecting stronger contraction in commuting for Work relative to Education.

Active and Public transport categories show more moderate and region-specific divergence, with both positive and negative values depending on local context.

Overall, the DiD results confirm that divergence between Education and Work is not spatially uniform and is most pronounced in metropolitan labour-market regions.

4.4.4 Urban versus Non-Urban Comparisons

To assess whether settlement structure moderates modal redistribution, five major urban regions (Auckland, Waikato, Bay of Plenty, Wellington, and Canterbury) and five non-urban regions (Gisborne, West Coast, Southland, Tasman, and Marlborough) were analysed separately. These classifications follow the grouping defined in Section 3.5.

4.4.4.1 Mode Shares in 2018 and 2023 (Top 5 Urban vs Non-Urban) The 2018-2023 comparison reveals clear structural contrasts between urban and non-urban contexts.

In **Urban regions**, Work travel remains heavily concentrated in Private transport (68.47% in 2023), but shows a pronounced shift toward At Home participation (10.13% to 17.91%). Education exhibits smaller changes, with a modest increase in At Home participation (4.65% to 7.61%) and relatively stable Private transport shares (49.80% to 50.57%).

In **Non-Urban regions**, Private transport dominates both journey types in both years, exceeding 73% for Work in 2023. Changes between 2018 and 2023 are comparatively muted particularly for Work indicating greater modal stability outside metropolitan contexts.

Urban regions experienced substantially larger increases in “At Home” participation for Work journeys relative to non-urban regions, while Education showed only marginally stronger increases in urban areas. Non-urban regions maintained consistently higher reliance on Private transport and exhibited more stable overall mode distributions.

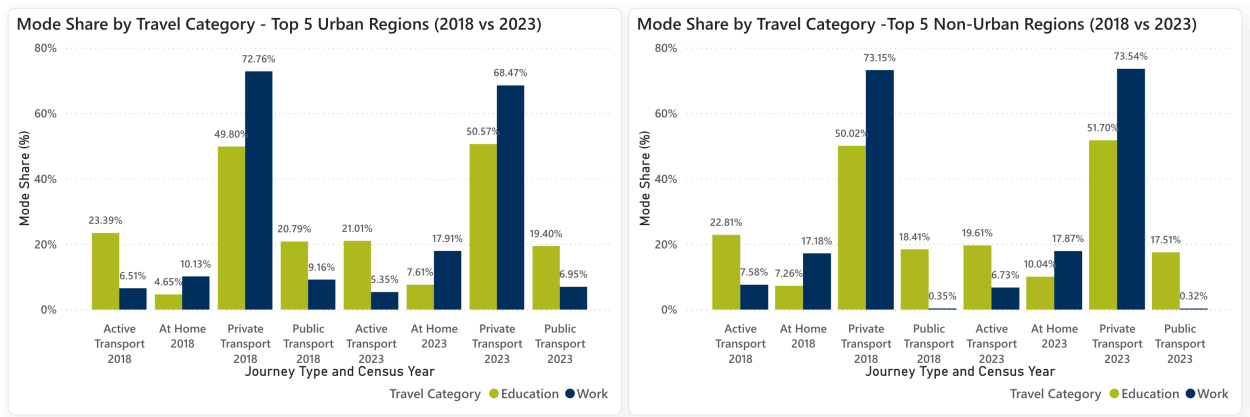


Figure 4: Mode Share by Travel Category - Top 5 Urban and Non-Urban Regions (2018 vs 2023)

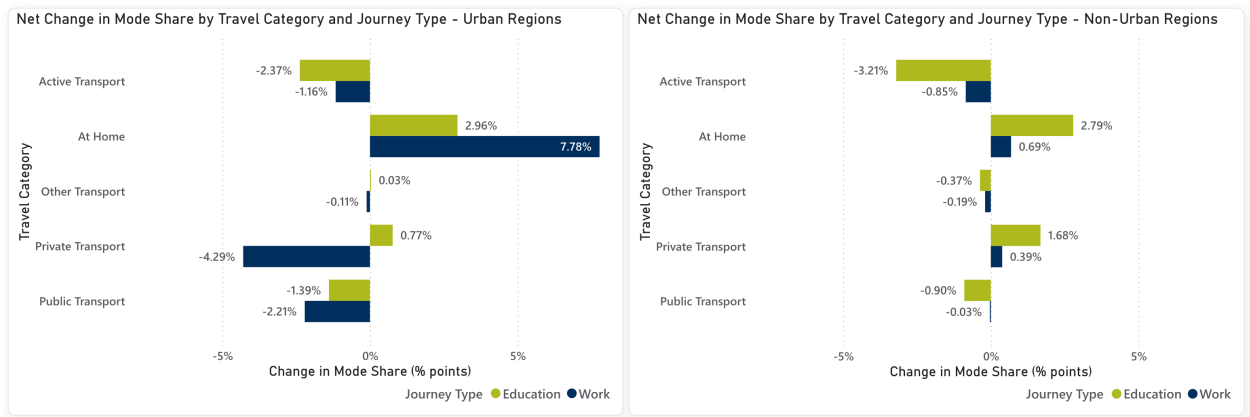


Figure 5: Net Change in Mode Share - Urban and Non-Urban Regions

4.4.4.2 Net Percentage-Point Changes (Urban vs Non-Urban)

Table 16: Travel mode shares (2018 and 2023) and percentage-point changes for Top 5 Urban and Non-Urban regions

Region Classification	Journey Type	Travel Category	2018 (%)	2023 (%)	Percentage-Point Change
Urban (Top 5)	Education	Active Transport	23.39	21.01	-2.37
Urban (Top 5)	Education	At Home	4.65	7.61	2.96
Urban (Top 5)	Education	Other Transport	1.38	1.41	0.03
Urban (Top 5)	Education	Private Transport	49.80	50.57	0.77
Urban (Top 5)	Education	Public Transport	20.79	19.40	-1.39
Urban (Top 5)	Work	Active Transport	6.51	5.35	-1.16
Urban (Top 5)	Work	At Home	10.13	17.91	7.78
Urban (Top 5)	Work	Other Transport	1.43	1.32	-0.11
Urban (Top 5)	Work	Private Transport	72.76	68.47	-4.29
Urban (Top 5)	Work	Public Transport	9.16	6.95	-2.21
Non-Urban (Top 5)	Education	Active Transport	22.81	19.61	-3.21
Non-Urban (Top 5)	Education	At Home	7.26	10.04	2.79
Non-Urban (Top 5)	Education	Other Transport	1.50	1.13	-0.37
Non-Urban (Top 5)	Education	Private Transport	50.02	51.70	1.68
Non-Urban (Top 5)	Education	Public Transport	18.41	17.51	-0.90
Non-Urban (Top 5)	Work	Active Transport	7.58	6.73	-0.85
Non-Urban (Top 5)	Work	At Home	17.18	17.87	0.69
Non-Urban (Top 5)	Work	Other Transport	1.74	1.54	-0.19
Non-Urban (Top 5)	Work	Private Transport	73.15	73.54	0.39
Non-Urban (Top 5)	Work	Public Transport	0.35	0.32	-0.03

The Net-change plots (Figure 5) and Table 16 indicate that modal reallocation between 2018 and 2023 was substantially more pronounced in urban regions than in non-urban regions.

In urban areas, Work journeys exhibit a marked behavioural shift. “At Home” participation increases sharply by +7.78 percentage points, accompanied by a substantial decline in Private transport (-4.29 pp) and reductions in Public transport (-2.21 pp) and Active transport (-1.16 pp). Education journeys in urban regions show more moderate adjustments, with a +2.96 pp increase in “At Home” participation, a small increase in Private transport (+0.77 pp), and modest declines in Active (-2.37 pp) and Public transport (-1.39 pp).

In contrast, non-urban regions display considerably smaller net changes, particularly for Work. “At Home” participation for Work increases only marginally (+0.69 pp), and Private transport remains effectively stable (+0.39 pp). Education in non-urban regions shows a moderate increase in “At Home” participation (+2.79 pp) and a rise in Private transport (+1.68 pp), but these adjustments are less dramatic than the urban Work shifts. Declines in Active transport are more pronounced for Education (-3.21 pp) than for Work (-0.85 pp), indicating some redistribution within education-related travel outside metropolitan areas.

Overall, the magnitude of modal adjustment is clearly larger in urban contexts, particularly for Work journeys. Non-urban regions demonstrate greater modal stability and continued reliance on Private transport, suggesting that structural change following 2018 was geographically uneven.

4.4.4.3 Difference-in-Differences: Education-Work Divergence

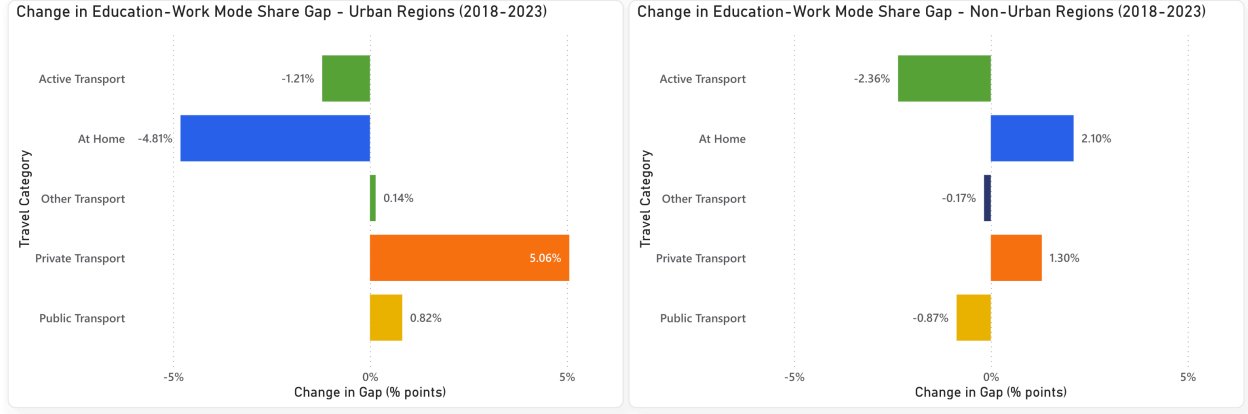


Figure 6: Change in Education-Work Mode Share Gap - Urban and Non-Urban Regions

Table 17: Difference-in-differences in percentage-point changes (2018–2023) between Education and Work travel mode shares for Top 5 Urban and Non-Urban regions

Urban Classification	Travel Category	Education Δ pp	Work Δ pp	Diff-in-Diff (Education – Work, pp)
Urban (Top 5)	Active Transport	-2.37	-1.16	-1.21
Urban (Top 5)	At Home	2.96	7.78	-4.82
Urban (Top 5)	Other Transport	0.03	-0.11	0.14
Urban (Top 5)	Private Transport	0.77	-4.29	5.06
Urban (Top 5)	Public Transport	-1.39	-2.21	0.82
Non-Urban (Top 5)	Active Transport	-3.21	-0.85	-2.36
Non-Urban (Top 5)	At Home	2.79	0.69	2.10
Non-Urban (Top 5)	Other Transport	-0.37	-0.19	-0.18
Non-Urban (Top 5)	Private Transport	1.68	0.39	1.29
Non-Urban (Top 5)	Public Transport	-0.90	-0.03	-0.87

The difference-in-differences (DiD) estimates quantify how the magnitude of temporal change differs between Education and Work journeys across region types. Positive values indicate relatively stronger growth (or weaker decline) for Education compared with Work, while negative values indicate relatively stronger change for Work.

In urban regions, divergence between journey purposes is pronounced. The DiD for “At Home” participation is strongly negative (-4.82 pp), indicating that remote participation increased substantially more for Work than for Education. Conversely, a large positive DiD in Private transport (+5.06 pp) reflects a sharper contraction in Work commuting relative to Education. Public transport shows moderate divergence (+0.82 pp), while Active transport exhibits a smaller negative DiD (-1.21 pp), suggesting slightly stronger decline in Education. Overall, the urban results reveal substantial reallocation concentrated in commuting-related modes.

In non-urban regions, DiD magnitudes are markedly smaller. The largest divergence occurs in Active transport (-2.36 pp), indicating a stronger decline for Education relative to Work. At-Home participation shows a positive DiD (+2.10 pp), reflecting relatively stronger growth in remote participation for Education than for Work. Divergence in Private transport (+1.29 pp) and Public transport (-0.87 pp) remains modest compared with urban patterns.

Taken together, the DiD analysis demonstrates that Education-Work divergence is substantially stronger in metropolitan contexts. Urban regions exhibit sharper differentiation in commuting-related modal adjustment, whereas non-urban regions show comparatively muted and more balanced changes between journey purposes.

4.4.4.4 Region-Specific Patterns

Figure 7 highlights substantial heterogeneity within both urban and non-urban classifications, indicating that divergence between Education and Work is not uniform across regions.

Among urban regions, the largest negative shifts in the At Home gap occur in Wellington (-6.68 pp) and Auckland (-6.36 pp), reflecting substantially stronger growth in remote participation for Work relative to Education in these metropolitan centres. Divergence in Private transport is also pronounced, exceeding +5 pp in Auckland (+5.96 pp) and Wellington (+5.42 pp), and remaining sizeable in Waikato (+3.70 pp) and Canterbury (+2.70 pp). Public transport divergence varies across regions, with Wellington showing a notable positive shift (+3.56 pp), while Canterbury (-2.03 pp) and Waikato (-1.10 pp) exhibit contraction. These patterns suggest that the largest metropolitan labour markets experienced the strongest reallocation in commuting-related modes.

Among non-urban regions, divergence is more heterogeneous and generally smaller in magnitude. The West Coast stands out with a strong positive shift in the At-Home gap (+5.39 pp), indicating relatively stronger Education growth in remote participation compared with Work. Gisborne displays the largest negative change in Active transport (-5.17 pp), signalling sharper contraction for Education. Private transport divergence varies moderately across regions, ranging from -2.58 pp in Marlborough to +4.92 pp in Gisborne. Overall, non-urban adjustments appear less systematic and more locally contingent.

Taken together, these region-specific results reinforce the broader finding that metropolitan areas exhibit stronger and more consistent Education-Work divergence, while non-urban regions demonstrate more uneven and context-specific patterns of modal change.

Regional Change in Education-Work Mode Share Gap - Top 5 Urban Regions					Regional Change in Education-Work Mode Share Gap - Top 5 Non-Urban Regions				
Values show change in the Education-Work mode share gap (% points), 2018-2023.					Values show change in the Education-Work mode share gap (% points), 2018-2023.				
Region	At Home	Private Transport	Public Transport	Active Transport	Region	At Home	Private Transport	Public Transport	Active Transport
Auckland Region	-6.36%	5.96%	1.19%	-1.01%	Gisborne Region	1.41%	4.92%	-0.96%	-5.17%
Bay of Plenty Region	-0.25%	1.53%	1.19%	-2.43%	Marlborough Region	2.02%	-2.58%	-0.06%	0.76%
Canterbury Region	-0.56%	2.70%	-2.03%	0.17%	Southland Region	2.81%	0.11%	-0.48%	-2.32%
Waikato Region	-0.70%	3.70%	-1.10%	-2.17%	Tasman Region	-0.27%	2.34%	-0.49%	-1.29%
Wellington Region	-6.68%	5.42%	3.56%	-2.52%	West Coast Region	5.39%	1.99%	-3.57%	-3.59%

Figure 7: Regional Change in Education-Work Mode Share Gap - Heatmap (Urban and Non-Urban)

4.4.4.5 Overall Findings for Top 5 Urban vs Non-Urban Regions Across all analyses, three consistent conclusions emerge:

1. **Urban regions experienced substantially stronger modal redistribution**, particularly in Work travel.
2. **Remote participation expanded most sharply in metropolitan labour-market centres**, widening the Education-Work gap.

3. **Non-Urban regions exhibit greater modal stability**, with smaller difference-in-differences magnitudes and more aligned shifts across journey types.

Settlement structure therefore plays a central moderating role in post-2018 travel behaviour change, with metropolitan regions driving the majority of structural divergence between Education and Work journeys.

4.4.5 City-Level Comparison

To further examine spatial heterogeneity, four representative territorial authorities were selected for focused analysis: two metropolitan centres (Wellington City and Christchurch City) and two non-metropolitan centres (Gisborne District and Invercargill City). These locations were chosen to contrast large urban labour-market hubs with smaller regional centres and to assess how travel-mode redistribution varied at a finer spatial scale beyond regional aggregation.

Percentage-point changes in travel mode shares (2018-2023) are reported in **Table A3 in the Appendix**, and the corresponding difference-in-differences (Education - Work) estimates are presented in **Table A4 (Appendix)**.

Percentage-Point Changes

Clear contrasts emerge across the selected cities.

In **Wellington City**, Work travel exhibits a substantial increase in *At Home* participation (+10.56 pp), accompanied by a marked decline in *Private transport* (-5.71 pp). Education also shows increased At Home participation (+3.79 pp) and growth in *Public transport* (+3.64 pp), while *Active transport* declines (-6.73 pp). These results indicate pronounced restructuring of commuting patterns in the capital.

In **Christchurch City**, changes are more moderate. Work At Home participation increases by +3.92 pp, while Private transport declines by -4.00 pp. Education shows smaller shifts, including a +2.88 pp increase in At Home participation and relatively stable modal shares elsewhere.

Among non-metropolitan centres, **Gisborne District** displays strong reductions in Education *Active transport* (-6.34 pp) and a notable increase in Education *Private transport* (+5.24 pp), suggesting modal substitution toward private vehicles. Work travel in Gisborne shows comparatively limited change.

In **Invercargill City**, modal adjustments are modest overall. Education *Active transport* declines (-3.89 pp), while At-Home participation increases moderately for both journey types (+2.70 pp for Education; +0.59 pp for Work). Private transport remains largely stable.

Difference-in-Differences (Education - Work)

Difference-in-differences estimates in **Table A4** formalise these contrasts.

Wellington City exhibits the largest divergence between journey types. The DiD for *At-Home* is -6.77 pp, indicating substantially stronger remote-work expansion relative to education. Strong positive DiD values in *Private transport* (+4.90 pp) and *Public transport* (+4.97 pp) reflect sharper contraction in commuting relative to education travel.

Christchurch City shows moderate divergence, particularly in *Private transport* (+2.94 pp), consistent with stronger commuting contraction for Work.

In Gisborne District, divergence is concentrated in *Private transport* (+4.92 pp) and *Active transport* (-5.17 pp), indicating education-specific shifts toward private modes. Invercargill City displays relatively small DiD magnitudes across categories, consistent with greater modal stability.

Summary

City-level evidence reinforces earlier regional findings:

- Metropolitan centres, particularly Wellington City, experienced the most pronounced divergence between Education and Work travel.
- Remote participation expanded far more strongly for Work than Education in large urban labour markets.
- Smaller regional centres exhibit more moderate and heterogeneous modal adjustments.

These results confirm that behavioural change is not only regionally differentiated but also strongly conditioned by local urban structure at the district/city level.

4.4.6 Change by Age Group (Education vs Work)

This subsection examines age-specific differences in modal restructuring between 2018 and 2023. We first describe mode shares, followed by net percentage-point changes, and finally Education-Work divergence (difference-in-differences).

The Under-15 age group is included for Education journeys but is structurally excluded from Work journey comparisons, as work-related travel is not defined for this cohort in the Census data. Consequently, difference-in-differences estimates are reported only for the 15-29, 30-64, and Over 64 age groups.

4.4.6.1 Mode Shares by Age Group

Mode shares for Education and Work across age groups are shown in *Figure 8*.

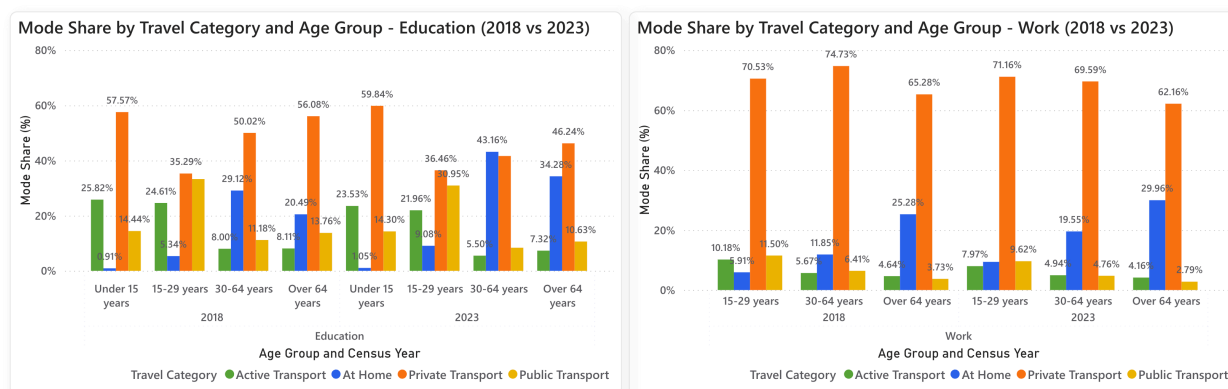


Figure 8: Mode share by travel category and age group - Education and Work (2018 vs 2023)

Modal structure varies systematically by age. Private transport remains the dominant mode across most age groups in both journey types, particularly among working-age adults.

For **Education**, younger groups show higher shares of Active and Public transport, whereas older cohorts rely more heavily on Private transport. By 2023, At-Home participation increases across all adult groups, particularly among 30-64 and Over-64 individuals.

For **Work**, Private transport remains dominant in both years. However, At-Home participation rises noticeably by 2023, especially among 30-64 and Over-64 workers.

4.4.6.2 Net Change in Mode Share (2018-2023)

Percentage-point changes in mode shares by age group are reported in **Table A5 in the (Appendix)** and visualised in Figure 9.

For Education, modal reallocation is concentrated among adult groups. The 30-64 age group exhibits the largest shift toward At Home participation (+14.04 percentage points), accompanied by substantial declines in Private transport (-8.38 pp), Public transport (-2.80 pp), and Active transport (-2.50 pp). A similar pattern is observed among those aged Over 64, where At Home participation increases markedly (+13.79 pp) and Private transport declines sharply (-9.84 pp). In contrast, the 15-29 age group shows more moderate change, with At Home rising by +3.73 pp and smaller reductions in Active (-2.65 pp) and Public transport (-2.34 pp). Under 15 groups display minimal redistribution overall.

For Work, directional patterns are broadly similar but magnitudes are smaller. Among 30-64 workers, At-Home participation increases by +7.69 pp, while Private transport declines by -5.14 pp. For those aged Over 64, At Home rises by +4.69 pp and Private transport decreases by -3.12 pp. The 15-29 group shows moderate At Home growth (+3.48 pp) and relatively minor adjustments across other modes.

Overall, age-based analysis reveals that the strongest modal reallocation occurred among prime working-age adults in Education, with Work exhibiting comparable but more moderate adjustments. Younger cohorts experienced smaller structural shifts, indicating that post-2018 redistribution was most concentrated among adult populations.

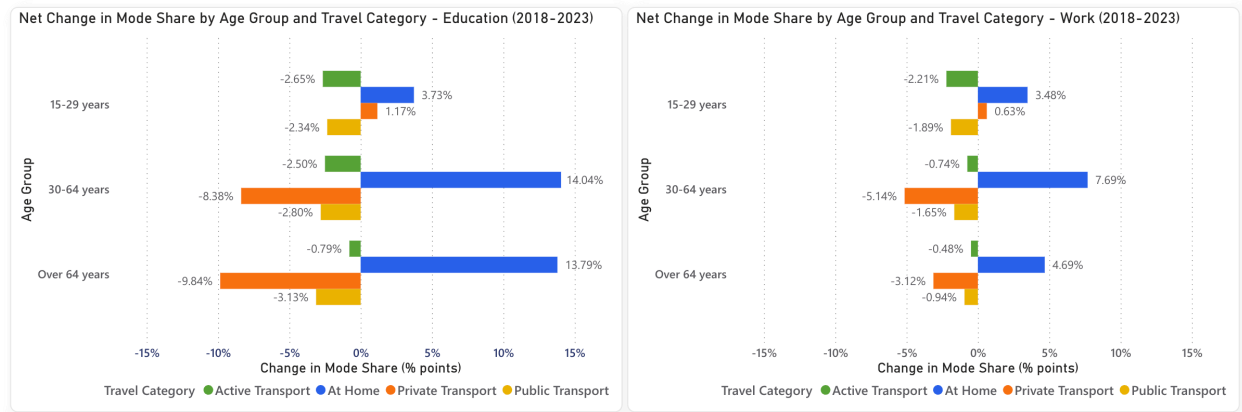


Figure 9: Net Change in Education-Work Mode Share - Age Group and Travel Category

4.4.6.3 Change in the Education-Work Gap

Education-Work divergence by age group is illustrated in Figure 10, with numerical values reported in **Table A6 in the (Appendix)**.

Difference-in-differences estimates indicate that divergence increases systematically with age. Among the 30-64 cohort, the At Home gap widens substantially (+6.35 percentage points), reflecting stronger growth in remote participation for Education relative to Work. Concurrently, Private transport shows a negative divergence (-3.24 pp), indicating a sharper contraction in Education than in Work. Declines in Public (-1.15 pp) and Active transport (-1.76 pp) are more moderate.

The largest divergence is observed among those aged Over 64. The At Home gap expands markedly (+9.10 pp), representing the strongest Education relative growth across all age groups. At the same time, Private transport divergence is pronounced (-6.73 pp), confirming that reductions in Private transport were substantially stronger for Education among older cohorts. Public transport (-2.19 pp) also shows notable contraction relative to Work.

In contrast, divergence among the 15-29 age group remains limited. Differences across all categories are small in magnitude (generally below one percentage point in absolute terms), indicating broadly parallel temporal adjustments between Education and Work for younger adults.

Overall, the age-specific analysis demonstrates that Education-Work divergence intensifies with age. The most substantial restructuring occurs among 30-64 and Over 64 cohorts, where Education experiences stronger relative expansion in At Home participation and sharper contraction in Private transport compared with Work. Younger cohorts exhibit comparatively stable Education-Work differentials.

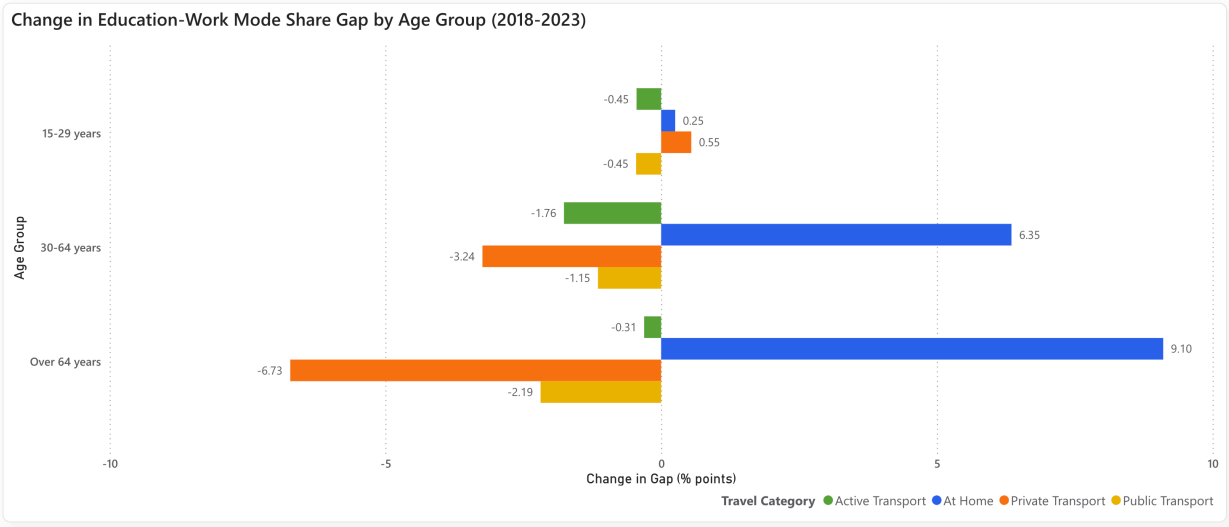


Figure 10: Change in Education-Work Mode Share Gap - Age Group and Travel Category

Summary

Behavioural restructuring is strongly age-conditioned.

The largest shifts occur among **30-64 and Over-64 cohorts**, where At-Home participation expands and Private transport contracts. Education exhibits stronger percentage-point growth in at-home participation than Work among older adults, producing widening Education-Work gaps.

5 Discussion

The objective of this study was to assess whether travel-mode distributions in New Zealand changed between Census 2018 and Census 2023, and whether those temporal shifts differed systematically between Education and Work journeys. Using a structured framework combining mode-share analysis, percentage-point change measurement, difference-in-differences comparison, distributional testing, and multinomial logistic modelling, the findings demonstrate clear evidence of structural modal redistribution over the post-pandemic period, with important variation across geography and age groups.

At the national level, both Education and Work journeys exhibit statistically significant changes in travel-mode composition between 2018 and 2023. However, the magnitude and pattern of these shifts differ across journey types. The most pronounced structural change occurs in the growth of *At Home* participation, particularly for Work journeys. The multinomial interaction model confirms that the temporal effect of Census year differs significantly between Education and Work journeys, providing formal statistical support for differential post-pandemic adjustment.

While both journey types experienced an expansion in remote participation, the increase is substantially larger for Work. This divergence suggests that structural remote working arrangements have persisted beyond the immediate pandemic period, whereas Education travel exhibits a comparatively smaller shift toward remote participation. The interaction effects estimated within the multinomial framework reinforce the conclusion that temporal change is not uniform across journey contexts.

The regional and sub-population extensions further clarify how national-level restructuring is spatially and demographically distributed. Metropolitan regions exhibit larger increases in *At Home* participation and sharper contractions in Private transport use relative to non-urban regions. This pattern is particularly pronounced for Work journeys, indicating that remote working capacity and modal substitution are more feasible in large urban labour markets. In contrast, non-urban regions display greater modal stability and continued reliance on Private transport, reflecting structural constraints in transport infrastructure and occupational composition.

City-level comparisons reinforce this spatial heterogeneity. Larger metropolitan centres demonstrate stronger modal redistribution, while smaller regional centres show comparatively muted change. These findings indicate that aggregate national trends are not geographically uniform but instead reflect concentrated behavioural shifts within specific spatial contexts.

Age-group analysis reveals that modal restructuring is also strongly life-stage dependent. The largest changes occur among the 30-64 and Over-64 cohorts, where *At Home* participation increases sharply and Private transport declines substantially. Younger cohorts display more moderate and parallel changes between Education and Work journeys. The widening Education-Work divergence among older adults suggests that remote working arrangements are disproportionately concentrated among prime-age and older workers, whereas education-related travel among younger populations remains more dependent on physical attendance.

Taken together, the results suggest that the post-2018 evolution of travel behaviour in New Zealand reflects a structural reconfiguration of commuting patterns rather than a temporary disruption. The persistence of remote work, particularly in metropolitan regions and among working-age adults, implies enduring implications for peak-hour congestion, transport investment planning, and public transport demand modelling.

The integration of multinomial modelling with descriptive extensions strengthens the interpretation of these findings. The regression framework provides a unified inferential test of interaction effects, while predicted probabilities translate statistical results into interpretable modal share differences. The complementary stratified analyses demonstrate how aggregate change emerges from heterogeneous demographic and spatial processes.

5.1 Limitations

Several limitations should be acknowledged when interpreting these findings.

First, the analysis relies on aggregated Census tabulations rather than individual-level microdata. While the multinomial count formulation is statistically appropriate for grouped outcomes, it precludes adjustment for additional covariates such as income, occupation, household structure, or car ownership. As a result, the observed shifts represent distributional change across populations rather than causal behavioural responses at the individual level.

Second, Census data capture the *main* means of travel on Census day rather than habitual multi-modal behaviour. Hybrid commuting patterns or occasional travel modes may therefore be underrepresented. In particular, the rapid adoption of hybrid work arrangements may not be fully reflected in a single-day snapshot.

Third, the urban versus non-urban grouping is necessarily a simplification. Although designed to contrast metropolitan and non-metropolitan contexts, regional classification cannot fully capture within-region heterogeneity or finer-grained local transport dynamics.

Fourth, A further limitation arises from the structure of the publicly available Census tabulations. The dataset does not provide linked information on residential location and specific workplace or educational institution addresses, nor does it include trip distance or travel time. As a result, the analysis cannot assess how spatial separation between origin and destination influences modal choice or remote participation.

Fifth, attempts were made to incorporate additional contextual datasets, including the Equity Index, Deprivation Index, and urban–rural classification layers. However, mismatched spatial resolutions (e.g., meshblock versus SA2), lack of one-to-one correspondence, and missing data limited their suitability for integration within the project scope. The inability to incorporate these contextual variables restricts assessment of socioeconomic and fine-grained spatial drivers of modal change.

Sixth, the analysis spans only two Census waves. Although the 2018-2023 comparison is uniquely informative due to the pandemic context, longer-term trend analysis would require additional temporal observations to distinguish cyclical variation from structural transition.

Finally, the study focuses exclusively on Education and Work journeys. Other travel purposes, such as shopping or recreational travel, may exhibit different patterns of modal adjustment and are not examined here.

Despite these limitations, the combination of large-scale national Census data, formal interaction modelling, and systematic stratified extensions provides a robust descriptive and inferential account of post-pandemic modal restructuring in New Zealand.

5.2 Conclusion and Future Work

This study provides comprehensive evidence that travel-mode distributions in New Zealand changed significantly between 2018 and 2023 and that these changes differ systematically between Education and Work journeys. The staged analytical design, progressing from share-based composition analysis to contingency-table testing and finally to multinomial interaction modelling, ensures that both descriptive magnitude and formal statistical interaction are evaluated. Collectively, the inferential framework establishes not only that modal distributions evolved over time, but that the magnitude and direction of change differ structurally between Education and Work journeys. The multinomial interaction model confirms that temporal shifts are not homogeneous across journey types. In particular, Work journeys exhibit substantially stronger growth in *At Home* participation relative to Education journeys.

Regional and demographic extensions reveal that these changes are unevenly distributed. Metropolitan regions and working-age adults account for a disproportionate share of modal restructuring, while non-urban regions and younger cohorts display comparatively greater stability. The Education-Work gap widens most strongly in urban labour markets and among older age groups, indicating that remote work adoption is more strongly concentrated within specific spatial and life-stage segments of the population.

These findings carry important implications for transport policy and planning. Persistent reductions in physical commuting may alter long-term demand forecasts for public transport and road infrastructure. At the same time, education-related travel remains comparatively more anchored to physical attendance, suggesting that transport strategies must differentiate between journey purposes rather than treating them as behaviourally equivalent.

Future research could extend this analysis in several directions. First, incorporating additional Census waves would allow assessment of whether observed changes represent permanent structural shifts or medium-term adjustment. Second, microdata-based modelling could examine how occupation, income, household structure, and vehicle access mediate modal transition. Third, spatial econometric approaches could explore finer-grained geographic variation below the regional level. Finally, behavioural choice models could be developed to test substitution patterns among travel modes more explicitly.

In summary, the post-pandemic period has reshaped the relationship between work, education, and mobility in New Zealand. The persistence and uneven distribution of modal change underscore the importance of journey-specific and context-sensitive transport policy in an evolving mobility landscape.

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Appendix A - R Code

Table A1: Regional mode shares (%) in 2018 and 2023 and percentage-point change (2023 – 2018): Education

Region	Journey Type	Travel Category	2018 (%)	2023 (%)	PP change (2023–2018)
Area Outside Region	Education	Active Transport	25.64	17.65	-7.99
Area Outside Region	Education	At Home	10.26	8.82	-1.43
Area Outside Region	Education	Other Transport	5.13	0.00	-5.13
Area Outside Region	Education	Private Transport	20.51	32.35	11.84
Area Outside Region	Education	Public Transport	38.46	41.18	2.71
Auckland Region	Education	Active Transport	22.47	20.33	-2.14
Auckland Region	Education	At Home	3.97	6.99	3.02
Auckland Region	Education	Other Transport	1.32	1.42	0.10
Auckland Region	Education	Private Transport	49.78	50.63	0.85
Auckland Region	Education	Public Transport	22.45	20.63	-1.83
Bay of Plenty Region	Education	Active Transport	21.17	17.88	-3.29
Bay of Plenty Region	Education	At Home	6.12	8.90	2.77
Bay of Plenty Region	Education	Other Transport	1.19	1.03	-0.16
Bay of Plenty Region	Education	Private Transport	53.22	52.82	-0.40
Bay of Plenty Region	Education	Public Transport	18.30	19.38	1.08
Canterbury Region	Education	Active Transport	26.78	26.38	-0.40
Canterbury Region	Education	At Home	5.51	8.22	2.72
Canterbury Region	Education	Other Transport	1.81	1.69	-0.11
Canterbury Region	Education	Private Transport	50.56	50.47	-0.09
Canterbury Region	Education	Public Transport	15.35	13.23	-2.12
Gisborne Region	Education	Active Transport	20.64	14.30	-6.34
Gisborne Region	Education	At Home	6.02	8.78	2.76
Gisborne Region	Education	Other Transport	1.85	1.33	-0.52
Gisborne Region	Education	Private Transport	55.19	60.43	5.24
Gisborne Region	Education	Public Transport	16.30	15.16	-1.14
Hawke's Bay Region	Education	Active Transport	23.41	18.99	-4.41
Hawke's Bay Region	Education	At Home	6.52	9.22	2.69
Hawke's Bay Region	Education	Other Transport	1.54	1.52	-0.01
Hawke's Bay Region	Education	Private Transport	54.06	57.47	3.42
Hawke's Bay Region	Education	Public Transport	14.47	12.79	-1.68
Manawatu-Wanganui Region	Education	Active Transport	22.21	17.82	-4.39
Manawatu-Wanganui Region	Education	At Home	7.75	10.44	2.69
Manawatu-Wanganui Region	Education	Other Transport	1.35	1.28	-0.07
Manawatu-Wanganui Region	Education	Private Transport	53.70	57.76	4.06
Manawatu-Wanganui Region	Education	Public Transport	14.98	12.69	-2.29
Marlborough Region	Education	Active Transport	25.61	25.05	-0.56
Marlborough Region	Education	At Home	8.85	11.93	3.08
Marlborough Region	Education	Other Transport	1.94	1.34	-0.60
Marlborough Region	Education	Private Transport	50.86	48.97	-1.89
Marlborough Region	Education	Public Transport	12.74	12.71	-0.03
Nelson Region	Education	Active Transport	37.09	34.25	-2.84
Nelson Region	Education	At Home	6.77	10.39	3.62
Nelson Region	Education	Other Transport	2.23	2.18	-0.05
Nelson Region	Education	Private Transport	45.00	43.79	-1.21
Nelson Region	Education	Public Transport	8.90	9.39	0.49
Northland Region	Education	Active Transport	13.18	10.23	-2.94

Table A1: Regional mode shares (%) in 2018 and 2023 and percentage-point change (2023 – 2018): Education (*continued*)

Region	Journey Type	Travel Category	2018 (%)	2023 (%)	PP change (2023–2018)
Northland Region	Education	At Home	7.29	10.24	2.94
Northland Region	Education	Other Transport	0.92	0.74	-0.17
Northland Region	Education	Private Transport	51.70	53.73	2.03
Northland Region	Education	Public Transport	26.91	25.05	-1.86
Otago Region	Education	Active Transport	37.64	35.87	-1.76
Otago Region	Education	At Home	5.63	8.08	2.45
Otago Region	Education	Other Transport	1.07	0.90	-0.17
Otago Region	Education	Private Transport	43.43	42.46	-0.96
Otago Region	Education	Public Transport	12.24	12.68	0.44
Southland Region	Education	Active Transport	21.11	18.24	-2.87
Southland Region	Education	At Home	6.05	8.59	2.54
Southland Region	Education	Other Transport	1.21	0.98	-0.23
Southland Region	Education	Private Transport	52.20	53.23	1.03
Southland Region	Education	Public Transport	19.43	18.96	-0.47
Taranaki Region	Education	Active Transport	21.07	17.76	-3.31
Taranaki Region	Education	At Home	8.09	10.58	2.49
Taranaki Region	Education	Other Transport	1.32	1.16	-0.16
Taranaki Region	Education	Private Transport	53.74	55.24	1.50
Taranaki Region	Education	Public Transport	15.79	15.26	-0.52
Tasman Region	Education	Active Transport	24.98	23.46	-1.52
Tasman Region	Education	At Home	8.47	10.77	2.30
Tasman Region	Education	Other Transport	1.59	1.28	-0.31
Tasman Region	Education	Private Transport	43.33	43.42	0.08
Tasman Region	Education	Public Transport	21.63	21.08	-0.55
Waikato Region	Education	Active Transport	21.15	17.78	-3.37
Waikato Region	Education	At Home	5.86	8.54	2.68
Waikato Region	Education	Other Transport	1.08	1.04	-0.04
Waikato Region	Education	Private Transport	55.61	57.49	1.87
Waikato Region	Education	Public Transport	16.29	15.15	-1.14
Wellington Region	Education	Active Transport	28.54	23.85	-4.68
Wellington Region	Education	At Home	5.98	9.26	3.27
Wellington Region	Education	Other Transport	1.64	1.60	-0.04
Wellington Region	Education	Private Transport	41.98	42.41	0.43
Wellington Region	Education	Public Transport	21.86	22.88	1.01
West Coast Region	Education	Active Transport	25.19	19.94	-5.25
West Coast Region	Education	At Home	9.47	13.79	4.32
West Coast Region	Education	Other Transport	1.06	0.69	-0.37
West Coast Region	Education	Private Transport	43.03	47.91	4.88
West Coast Region	Education	Public Transport	21.25	17.68	-3.57

Table A2: Regional mode shares (%) in 2018 and 2023 and percentage-point change (2023 – 2018): Work

Region	Journey Type	Travel Category	2018 (%)	2023 (%)	PP change (2023–2018)
Area Outside Region	Work	Active Transport	7.53	7.14	-0.39
Area Outside Region	Work	At Home	20.55	27.14	6.59
Area Outside Region	Work	Other Transport	4.11	2.86	-1.25
Area Outside Region	Work	Private Transport	66.44	62.86	-3.58
Area Outside Region	Work	Public Transport	1.37	0.00	-1.37
Auckland Region	Work	Active Transport	5.37	4.24	-1.13
Auckland Region	Work	At Home	8.77	18.15	9.37
Auckland Region	Work	Other Transport	1.32	1.21	-0.12
Auckland Region	Work	Private Transport	73.76	68.65	-5.11

Table A2: Regional mode shares (%) in 2018 and 2023 and percentage-point change (2023 – 2018): Work (*continued*)

Region	Journey Type	Travel Category	2018 (%)	2023 (%)	PP change (2023–2018)
Auckland Region	Work	Public Transport	10.78	7.76	-3.02
Bay of Plenty Region	Work	Active Transport	5.12	4.25	-0.87
Bay of Plenty Region	Work	At Home	14.47	17.50	3.03
Bay of Plenty Region	Work	Other Transport	1.29	1.16	-0.13
Bay of Plenty Region	Work	Private Transport	77.94	76.02	-1.92
Bay of Plenty Region	Work	Public Transport	1.18	1.07	-0.11
Canterbury Region	Work	Active Transport	8.06	7.49	-0.56
Canterbury Region	Work	At Home	12.41	15.69	3.28
Canterbury Region	Work	Other Transport	1.51	1.67	0.16
Canterbury Region	Work	Private Transport	75.17	72.38	-2.78
Canterbury Region	Work	Public Transport	2.85	2.76	-0.09
Gisborne Region	Work	Active Transport	6.06	4.88	-1.17
Gisborne Region	Work	At Home	15.05	16.40	1.35
Gisborne Region	Work	Other Transport	1.51	1.19	-0.32
Gisborne Region	Work	Private Transport	76.93	77.25	0.32
Gisborne Region	Work	Public Transport	0.45	0.27	-0.18
Hawke's Bay Region	Work	Active Transport	5.86	4.67	-1.20
Hawke's Bay Region	Work	At Home	13.77	15.73	1.96
Hawke's Bay Region	Work	Other Transport	1.38	0.97	-0.41
Hawke's Bay Region	Work	Private Transport	78.36	78.17	-0.19
Hawke's Bay Region	Work	Public Transport	0.62	0.47	-0.15
Manawatu-Whanganui Region	Work	Active Transport	7.87	5.91	-1.95
Manawatu-Whanganui Region	Work	At Home	13.92	16.34	2.42
Manawatu-Whanganui Region	Work	Other Transport	1.70	1.36	-0.34
Manawatu-Whanganui Region	Work	Private Transport	75.47	75.59	0.12
Manawatu-Whanganui Region	Work	Public Transport	1.04	0.79	-0.24
Marlborough Region	Work	Active Transport	9.17	7.84	-1.32
Marlborough Region	Work	At Home	15.65	16.71	1.06
Marlborough Region	Work	Other Transport	1.42	0.96	-0.46
Marlborough Region	Work	Private Transport	73.60	74.30	0.69
Marlborough Region	Work	Public Transport	0.16	0.19	0.03
Nelson Region	Work	Active Transport	13.63	14.49	0.85
Nelson Region	Work	At Home	12.18	14.85	2.66
Nelson Region	Work	Other Transport	1.98	1.91	-0.07
Nelson Region	Work	Private Transport	71.21	67.55	-3.66
Nelson Region	Work	Public Transport	0.99	1.20	0.21
Northland Region	Work	Active Transport	4.71	3.76	-0.95
Northland Region	Work	At Home	17.22	20.72	3.50
Northland Region	Work	Other Transport	1.31	1.07	-0.24
Northland Region	Work	Private Transport	76.16	74.05	-2.11
Northland Region	Work	Public Transport	0.60	0.40	-0.20
Otago Region	Work	Active Transport	11.53	10.19	-1.34
Otago Region	Work	At Home	14.05	16.43	2.38
Otago Region	Work	Other Transport	1.17	1.10	-0.07
Otago Region	Work	Private Transport	70.60	69.05	-1.55
Otago Region	Work	Public Transport	2.64	3.22	0.58
Southland Region	Work	Active Transport	6.26	5.71	-0.55
Southland Region	Work	At Home	17.28	17.01	-0.27
Southland Region	Work	Other Transport	2.16	2.05	-0.11
Southland Region	Work	Private Transport	73.86	74.78	0.92
Southland Region	Work	Public Transport	0.44	0.45	0.01
Taranaki Region	Work	Active Transport	6.55	5.22	-1.32
Taranaki Region	Work	At Home	17.51	18.81	1.30
Taranaki Region	Work	Other Transport	2.19	1.75	-0.43

Table A2: Regional mode shares (%) in 2018 and 2023 and percentage-point change (2023 – 2018): Work (*continued*)

Region	Journey Type	Travel Category	2018 (%)	2023 (%)	PP change (2023–2018)
Taranaki Region	Work	Private Transport	73.32	73.75	0.43
Taranaki Region	Work	Public Transport	0.43	0.46	0.03
Tasman Region	Work	Active Transport	8.31	8.08	-0.23
Tasman Region	Work	At Home	18.27	20.83	2.57
Tasman Region	Work	Other Transport	1.55	1.53	-0.02
Tasman Region	Work	Private Transport	71.46	69.20	-2.26
Tasman Region	Work	Public Transport	0.42	0.36	-0.05
Waikato Region	Work	Active Transport	5.72	4.52	-1.20
Waikato Region	Work	At Home	14.97	18.34	3.37
Waikato Region	Work	Other Transport	1.69	1.38	-0.31
Waikato Region	Work	Private Transport	76.31	74.49	-1.82
Waikato Region	Work	Public Transport	1.31	1.27	-0.05
Wellington Region	Work	Active Transport	12.90	10.74	-2.17
Wellington Region	Work	At Home	9.10	19.05	9.95
Wellington Region	Work	Other Transport	1.85	1.59	-0.25
Wellington Region	Work	Private Transport	58.15	53.16	-4.99
Wellington Region	Work	Public Transport	18.01	15.46	-2.55
West Coast Region	Work	Active Transport	10.20	8.53	-1.67
West Coast Region	Work	At Home	20.30	19.22	-1.08
West Coast Region	Work	Other Transport	1.48	1.33	-0.15
West Coast Region	Work	Private Transport	67.90	70.79	2.89
West Coast Region	Work	Public Transport	0.12	0.12	0.00

Table A3: Travel mode shares and percentage-point changes (2018–2023) for selected districts

District	Journey Type	Travel Category	2018 (%)	2023 (%)	Percentage-Point Change (2018–2023)
Christchurch City	Education	Active Transport	30.10	29.79	-0.32
Christchurch City	Education	At Home	4.90	7.77	2.88
Christchurch City	Education	Other Transport	2.08	1.96	-0.11
Christchurch City	Education	Private Transport	51.27	50.21	-1.06
Christchurch City	Education	Public Transport	11.66	10.27	-1.38
Christchurch City	Work	Active Transport	9.59	9.61	0.02
Christchurch City	Work	At Home	9.09	13.02	3.92
Christchurch City	Work	Other Transport	1.28	1.55	0.27
Christchurch City	Work	Private Transport	75.74	71.74	-4.00
Christchurch City	Work	Public Transport	4.30	4.08	-0.22
Gisborne District	Education	Active Transport	20.64	14.30	-6.34
Gisborne District	Education	At Home	6.02	8.78	2.76
Gisborne District	Education	Other Transport	1.85	1.33	-0.52
Gisborne District	Education	Private Transport	55.19	60.43	5.24
Gisborne District	Education	Public Transport	16.30	15.16	-1.14
Gisborne District	Work	Active Transport	6.06	4.88	-1.17
Gisborne District	Work	At Home	15.05	16.40	1.35
Gisborne District	Work	Other Transport	1.51	1.19	-0.32
Gisborne District	Work	Private Transport	76.93	77.25	0.32
Gisborne District	Work	Public Transport	0.45	0.27	-0.18
Invercargill City	Education	Active Transport	23.60	19.72	-3.89
Invercargill City	Education	At Home	5.88	8.57	2.70
Invercargill City	Education	Other Transport	1.40	1.24	-0.16
Invercargill City	Education	Private Transport	60.86	61.49	0.62

Table A3: Travel mode shares and percentage-point changes (2018–2023)
for selected districts (*continued*)

District	Journey Type	Travel Category	2018 (%)	2023 (%)	Percentage-Point Change (2018–2023)
Invercargill City	Education	Public Transport	8.26	8.99	0.73
Invercargill City	Work	Active Transport	6.15	5.52	-0.63
Invercargill City	Work	At Home	8.64	9.22	0.59
Invercargill City	Work	Other Transport	1.20	1.09	-0.11
Invercargill City	Work	Private Transport	83.43	83.54	0.12
Invercargill City	Work	Public Transport	0.58	0.62	0.03
Wellington City	Education	Active Transport	38.21	31.48	-6.73
Wellington City	Education	At Home	5.44	9.22	3.79
Wellington City	Education	Other Transport	2.12	2.23	0.11
Wellington City	Education	Private Transport	30.58	29.77	-0.81
Wellington City	Education	Public Transport	23.66	27.30	3.64
Wellington City	Work	Active Transport	23.27	19.78	-3.49
Wellington City	Work	At Home	8.11	18.66	10.56
Wellington City	Work	Other Transport	2.56	2.54	-0.02
Wellington City	Work	Private Transport	44.71	38.99	-5.71
Wellington City	Work	Public Transport	21.36	20.03	-1.33

Table A4: Difference-in-differences in percentage-point changes (2018–2023) between Education and Work travel mode shares for selected districts

District	Travel Category	Education Δ pp	Work Δ pp	Diff-in-Diff (Education – Work, pp)
Christchurch City	Active Transport	-0.32	0.02	-0.34
Christchurch City	At Home	2.88	3.92	-1.04
Christchurch City	Other Transport	-0.11	0.27	-0.38
Christchurch City	Private Transport	-1.06	-4.00	2.94
Christchurch City	Public Transport	-1.38	-0.22	-1.16
Gisborne District	Active Transport	-6.34	-1.17	-5.17
Gisborne District	At Home	2.76	1.35	1.41
Gisborne District	Other Transport	-0.52	-0.32	-0.20
Gisborne District	Private Transport	5.24	0.32	4.92
Gisborne District	Public Transport	-1.14	-0.18	-0.96
Invercargill City	Active Transport	-3.89	-0.63	-3.26
Invercargill City	At Home	2.70	0.59	2.11
Invercargill City	Other Transport	-0.16	-0.11	-0.05
Invercargill City	Private Transport	0.62	0.12	0.50
Invercargill City	Public Transport	0.73	0.03	0.70
Wellington City	Active Transport	-6.73	-3.49	-3.24
Wellington City	At Home	3.79	10.56	-6.77
Wellington City	Other Transport	0.11	-0.02	0.13
Wellington City	Private Transport	-0.81	-5.71	4.90
Wellington City	Public Transport	3.64	-1.33	4.97

Table A5: Percentage-point changes (2018–2023) in travel mode shares by age group and journey type

Journey Type	Age Group	Travel Category	Percentage-Point Change (2018–2023)
Education	15-29 years	At Home	3.73
Education	15-29 years	Active Transport	-2.65
Education	15-29 years	Public Transport	-2.34
Education	15-29 years	Private Transport	1.17
Education	15-29 years	Other Transport	0.08
Education	30-64 years	At Home	14.04
Education	30-64 years	Private Transport	-8.38
Education	30-64 years	Public Transport	-2.80
Education	30-64 years	Active Transport	-2.50
Education	30-64 years	Other Transport	-0.37
Education	Over 64 years	At Home	13.79
Education	Over 64 years	Private Transport	-9.84
Education	Over 64 years	Public Transport	-3.13
Education	Over 64 years	Active Transport	-0.79
Education	Over 64 years	Other Transport	-0.03
Education	Under 15 years	Active Transport	-2.29
Education	Under 15 years	Private Transport	2.27
Education	Under 15 years	At Home	0.14
Education	Under 15 years	Public Transport	-0.13
Education	Under 15 years	Other Transport	0.01
Work	15-29 years	At Home	3.48
Work	15-29 years	Active Transport	-2.21
Work	15-29 years	Public Transport	-1.89
Work	15-29 years	Private Transport	0.63
Work	15-29 years	Other Transport	-0.01
Work	30-64 years	At Home	7.69
Work	30-64 years	Private Transport	-5.14
Work	30-64 years	Public Transport	-1.65
Work	30-64 years	Active Transport	-0.74
Work	30-64 years	Other Transport	-0.17
Work	Over 64 years	At Home	4.69
Work	Over 64 years	Private Transport	-3.12
Work	Over 64 years	Public Transport	-0.94
Work	Over 64 years	Active Transport	-0.48
Work	Over 64 years	Other Transport	-0.15

Table A6: Difference-in-differences in percentage-point changes (2018–2023) between Education and Work travel mode shares by age group

Age Group	Travel Category	Education Δ pp	Work Δ pp	Diff-in-Diff (Education – Work, pp)
15-29 years	Private Transport	1.17	0.63	0.54
15-29 years	Public Transport	-2.34	-1.89	-0.45
15-29 years	Active Transport	-2.65	-2.21	-0.44
15-29 years	At Home	3.73	3.48	0.25
15-29 years	Other Transport	0.08	-0.01	0.09
30-64 years	At Home	14.04	7.69	6.35
30-64 years	Private Transport	-8.38	-5.14	-3.24
30-64 years	Active Transport	-2.50	-0.74	-1.76

Table A6: Difference-in-differences in percentage-point changes (2018–2023) between Education and Work travel mode shares by age group
(continued)

Age Group	Travel Category	Education Δ pp	Work Δ pp	Diff-in-Diff (Education – Work, pp)
30-64 years	Public Transport	-2.80	-1.65	-1.15
30-64 years	Other Transport	-0.37	-0.17	-0.20
Over 64 years	At Home	13.79	4.69	9.10
Over 64 years	Private Transport	-9.84	-3.12	-6.72
Over 64 years	Public Transport	-3.13	-0.94	-2.19
Over 64 years	Active Transport	-0.79	-0.48	-0.31
Over 64 years	Other Transport	-0.03	-0.15	0.12
Under 15 years	Active Transport	-2.29	NA	NA
Under 15 years	At Home	0.14	NA	NA
Under 15 years	Other Transport	0.01	NA	NA
Under 15 years	Private Transport	2.27	NA	NA
Under 15 years	Public Transport	-0.13	NA	NA

Table A7: Region-level percentage-point changes (2018–2023) and difference-in-differences between Education and Work travel mode shares

Region	Travel Category	Education Δ pp	Work Δ pp	Diff-in-Diff (Education – Work, pp)
Area Outside Region	Private Transport	11.84	-3.58	15.42
Area Outside Region	At Home	-1.43	6.59	-8.02
Area Outside Region	Active Transport	-7.99	-0.39	-7.60
Area Outside Region	Public Transport	2.71	-1.37	4.08
Area Outside Region	Other Transport	-5.13	-1.25	-3.88
Auckland Region	At Home	3.02	9.37	-6.35
Auckland Region	Private Transport	0.85	-5.11	5.96
Auckland Region	Public Transport	-1.83	-3.02	1.19
Auckland Region	Active Transport	-2.14	-1.13	-1.01
Auckland Region	Other Transport	0.10	-0.12	0.22
Bay of Plenty Region	Active Transport	-3.29	-0.87	-2.42
Bay of Plenty Region	Private Transport	-0.40	-1.92	1.52
Bay of Plenty Region	Public Transport	1.08	-0.11	1.19
Bay of Plenty Region	At Home	2.77	3.03	-0.26
Bay of Plenty Region	Other Transport	-0.16	-0.13	-0.03
Canterbury Region	Private Transport	-0.09	-2.78	2.69
Canterbury Region	Public Transport	-2.12	-0.09	-2.03
Canterbury Region	At Home	2.72	3.28	-0.56
Canterbury Region	Other Transport	-0.11	0.16	-0.27
Canterbury Region	Active Transport	-0.40	-0.56	0.16
Gisborne Region	Active Transport	-6.34	-1.17	-5.17
Gisborne Region	Private Transport	5.24	0.32	4.92
Gisborne Region	At Home	2.76	1.35	1.41
Gisborne Region	Public Transport	-1.14	-0.18	-0.96
Gisborne Region	Other Transport	-0.52	-0.32	-0.20
Hawke's Bay Region	Private Transport	3.42	-0.19	3.61
Hawke's Bay Region	Active Transport	-4.41	-1.20	-3.21
Hawke's Bay Region	Public Transport	-1.68	-0.15	-1.53
Hawke's Bay Region	At Home	2.69	1.96	0.73
Hawke's Bay Region	Other Transport	-0.01	-0.41	0.40

Table A7: Region-level percentage-point changes (2018–2023) and difference-in-differences between Education and Work travel mode shares
(continued)

Region	Travel Category	Education Δ pp	Work Δ pp	Diff-in-Diff (Education – Work, pp)
Manawatu-Whanganui Region	Private Transport	4.06	0.12	3.94
Manawatu-Whanganui Region	Active Transport	-4.39	-1.95	-2.44
Manawatu-Whanganui Region	Public Transport	-2.29	-0.24	-2.05
Manawatu-Whanganui Region	At Home	2.69	2.42	0.27
Manawatu-Whanganui Region	Other Transport	-0.07	-0.34	0.27
Marlborough Region	Private Transport	-1.89	0.69	-2.58
Marlborough Region	At Home	3.08	1.06	2.02
Marlborough Region	Active Transport	-0.56	-1.32	0.76
Marlborough Region	Other Transport	-0.60	-0.46	-0.14
Marlborough Region	Public Transport	-0.03	0.03	-0.06
Nelson Region	Active Transport	-2.84	0.85	-3.69
Nelson Region	Private Transport	-1.21	-3.66	2.45
Nelson Region	At Home	3.62	2.66	0.96
Nelson Region	Public Transport	0.49	0.21	0.28
Nelson Region	Other Transport	-0.05	-0.07	0.02
Northland Region	Private Transport	2.03	-2.11	4.14
Northland Region	Active Transport	-2.94	-0.95	-1.99
Northland Region	Public Transport	-1.86	-0.20	-1.66
Northland Region	At Home	2.94	3.50	-0.56
Northland Region	Other Transport	-0.17	-0.24	0.07
Otago Region	Private Transport	-0.96	-1.55	0.59
Otago Region	Active Transport	-1.76	-1.34	-0.42
Otago Region	Public Transport	0.44	0.58	-0.14
Otago Region	Other Transport	-0.17	-0.07	-0.10
Otago Region	At Home	2.45	2.38	0.07
Southland Region	At Home	2.54	-0.27	2.81
Southland Region	Active Transport	-2.87	-0.55	-2.32
Southland Region	Public Transport	-0.47	0.01	-0.48
Southland Region	Other Transport	-0.23	-0.11	-0.12
Southland Region	Private Transport	1.03	0.92	0.11
Taranaki Region	Active Transport	-3.31	-1.32	-1.99
Taranaki Region	At Home	2.49	1.30	1.19
Taranaki Region	Private Transport	1.50	0.43	1.07
Taranaki Region	Public Transport	-0.52	0.03	-0.55
Taranaki Region	Other Transport	-0.16	-0.43	0.27
Tasman Region	Private Transport	0.08	-2.26	2.34
Tasman Region	Active Transport	-1.52	-0.23	-1.29
Tasman Region	Public Transport	-0.55	-0.05	-0.50
Tasman Region	Other Transport	-0.31	-0.02	-0.29
Tasman Region	At Home	2.30	2.57	-0.27
Waikato Region	Private Transport	1.87	-1.82	3.69
Waikato Region	Active Transport	-3.37	-1.20	-2.17
Waikato Region	Public Transport	-1.14	-0.05	-1.09
Waikato Region	At Home	2.68	3.37	-0.69
Waikato Region	Other Transport	-0.04	-0.31	0.27
Wellington Region	At Home	3.27	9.95	-6.68
Wellington Region	Private Transport	0.43	-4.99	5.42
Wellington Region	Public Transport	1.01	-2.55	3.56

Table A7: Region-level percentage-point changes (2018–2023) and difference-in-differences between Education and Work travel mode shares
(*continued*)

Region	Travel Category	Education Δ pp	Work Δ pp	Diff-in-Diff (Education – Work, pp)
Wellington Region	Active Transport	-4.68	-2.17	-2.51
Wellington Region	Other Transport	-0.04	-0.25	0.21
West Coast Region	At Home	4.32	-1.08	5.40
West Coast Region	Active Transport	-5.25	-1.67	-3.58
West Coast Region	Public Transport	-3.57	0.00	-3.57
West Coast Region	Private Transport	4.88	2.89	1.99
West Coast Region	Other Transport	-0.37	-0.15	-0.22